Lecture 24 (11/11)

Regularization and orthogonal features

Last day of final material

Review: the idea of regularization

fitting is a minimization problem

$\hat{\beta}$ minimizes the training error

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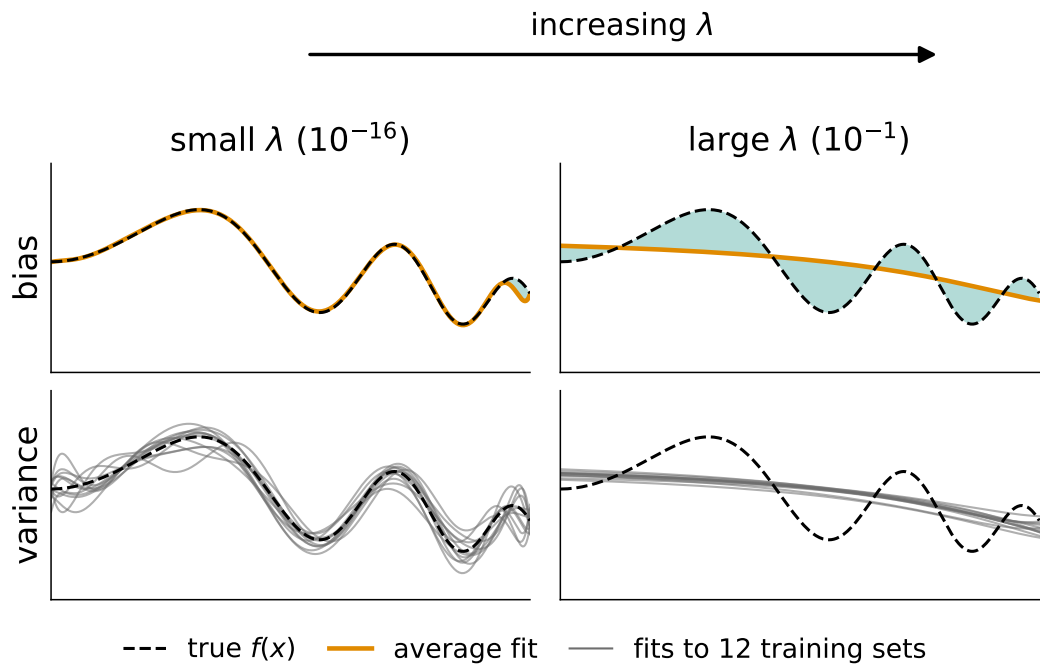
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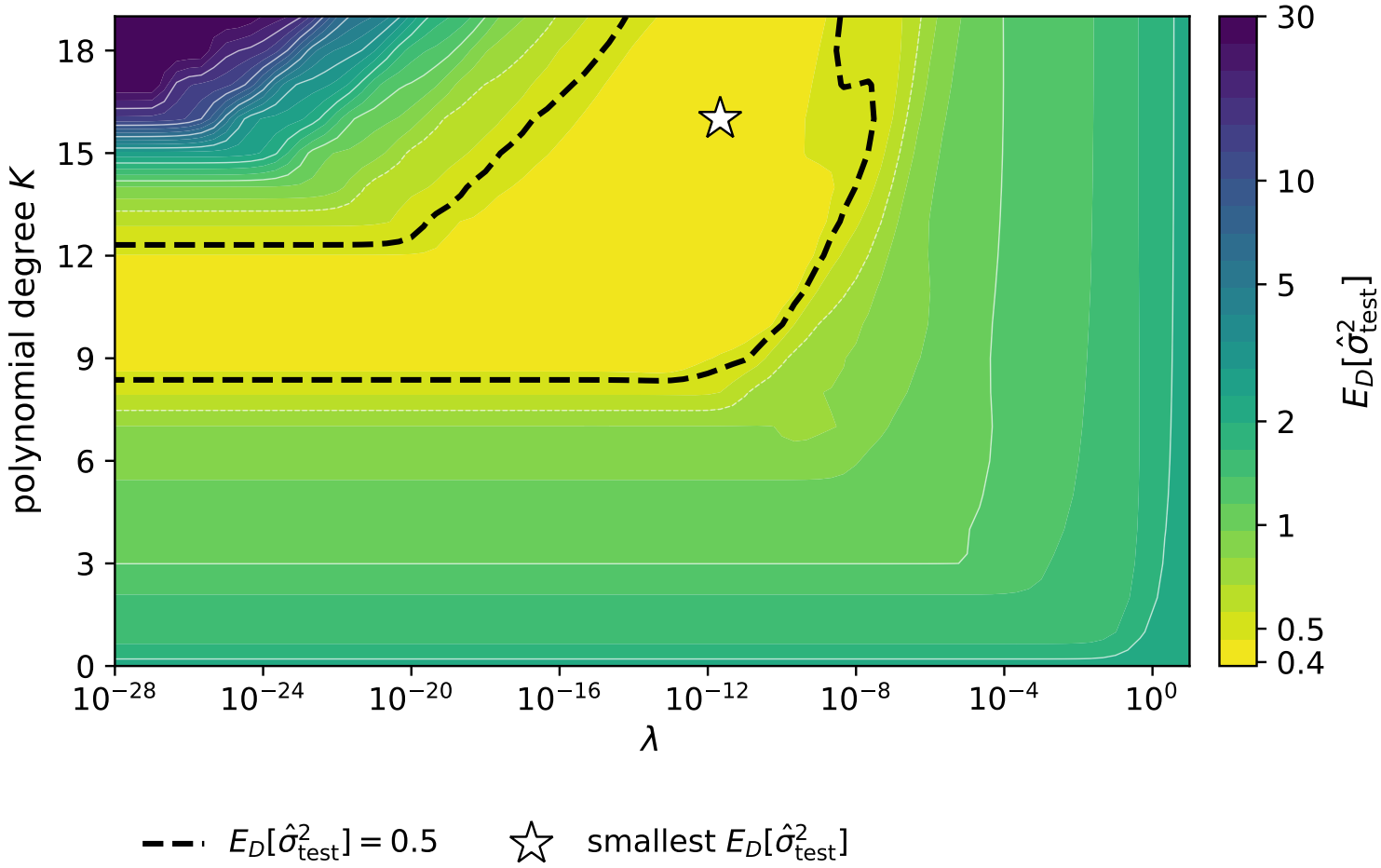
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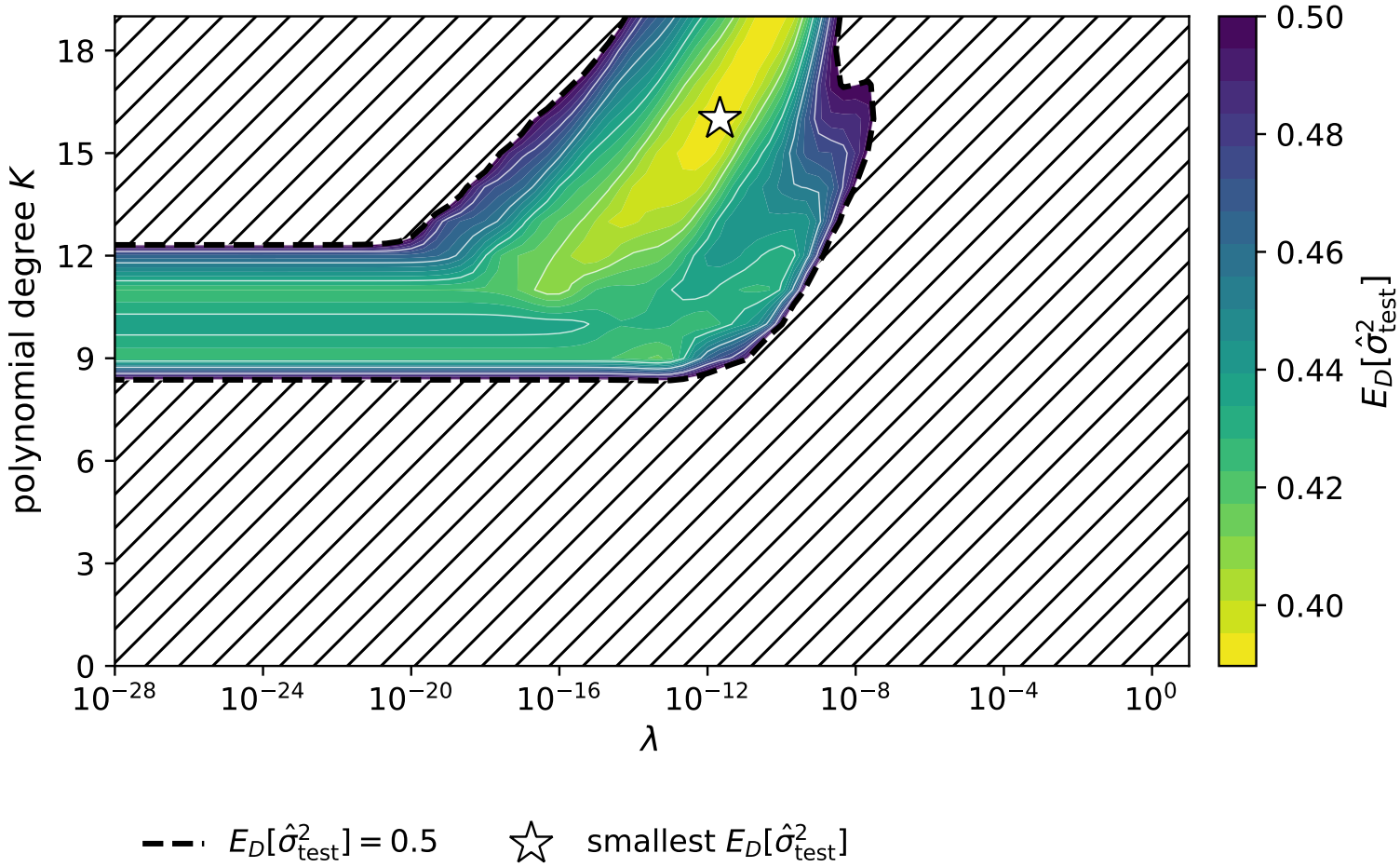
larger $\lambda \Rightarrow$ smaller coefficients: bias $\uparrow$, variance $\downarrow$



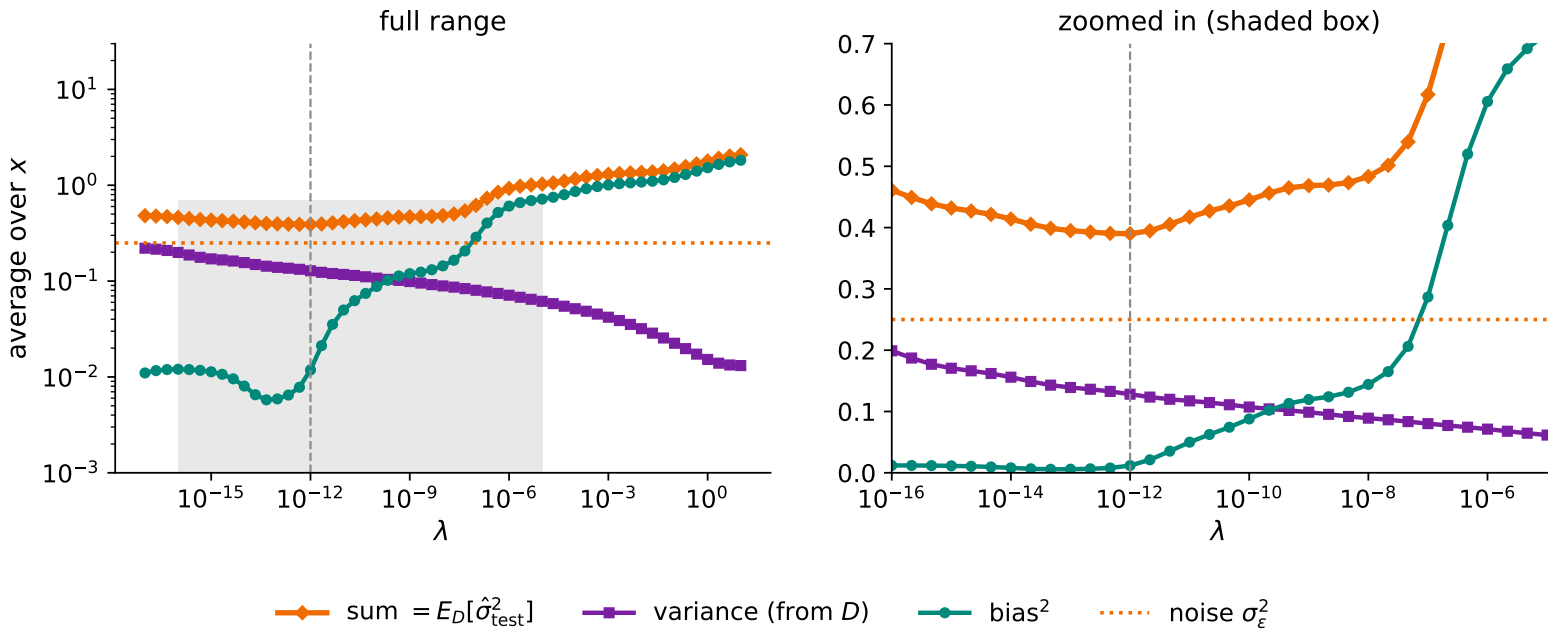
Review from Monday: Contour Plot



Review from Monday: Contour Plot



Review from Monday: Bias and variance vs. λ



The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

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training set

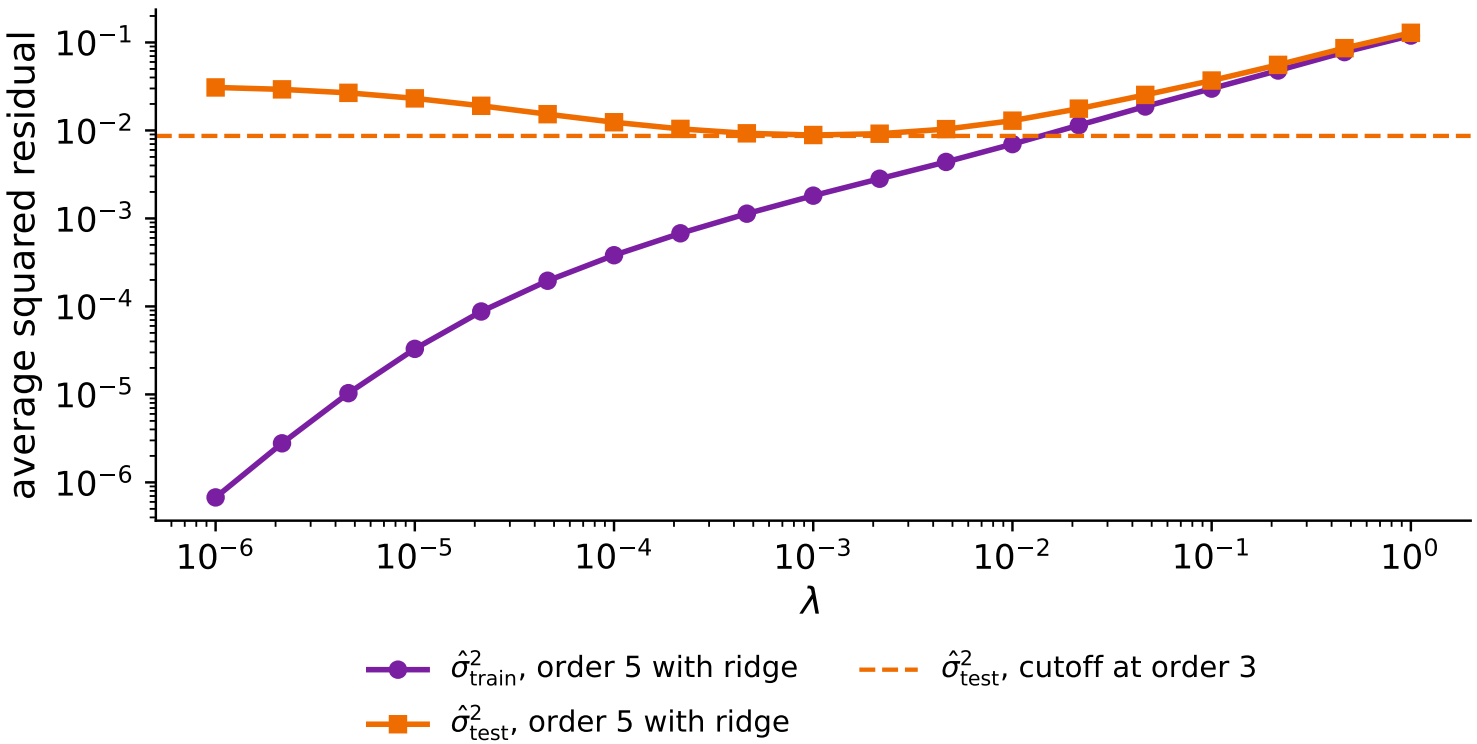
$N_{\text{train}} = 4000$ variants

fit with ridge regression

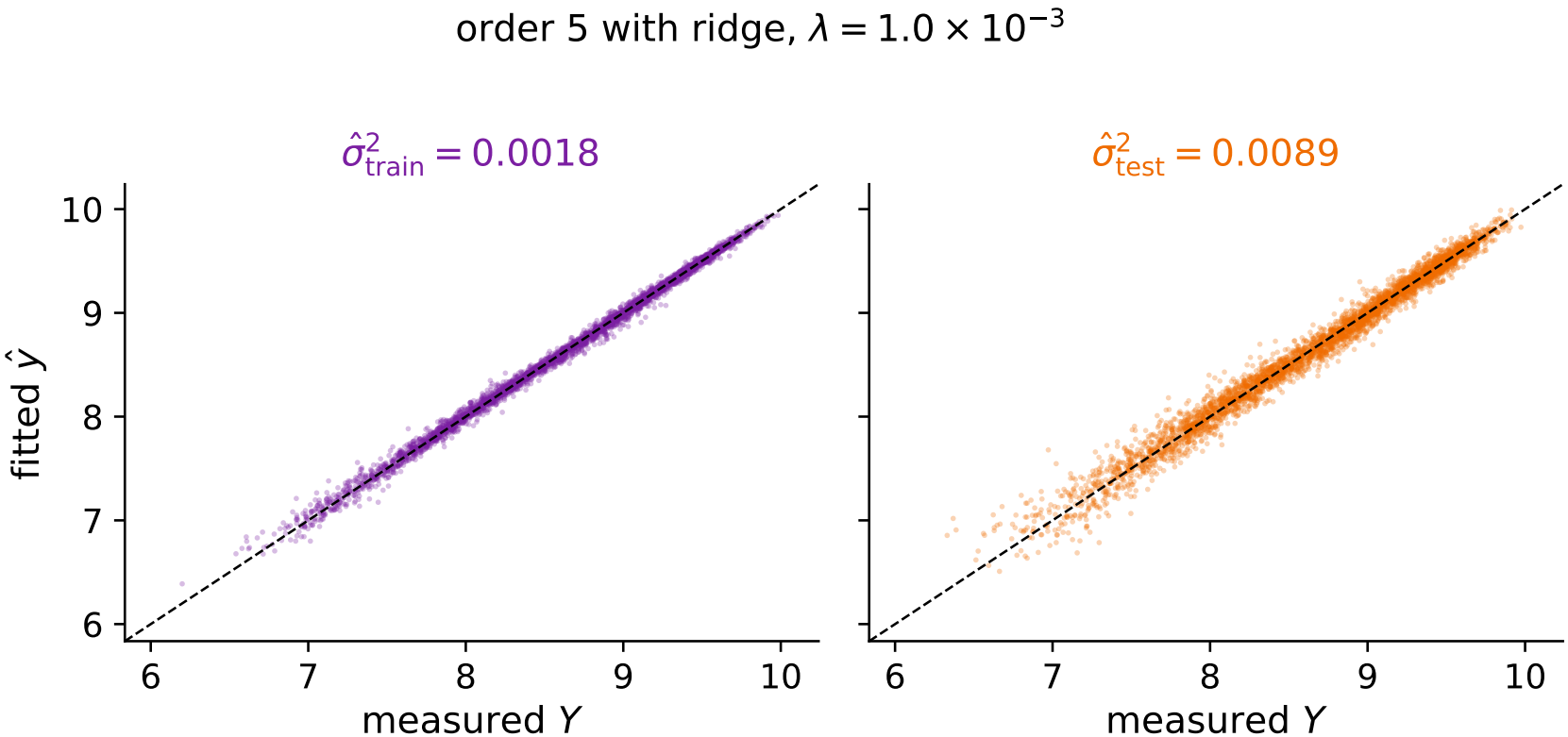
minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

Regularization for the spike protein

All interactions up to order 5 (4944 coefficients) fit with ridge regression with the to the same 4000 training variants as on 11/6 slide. The dashed line is least squares with the cutoff at order 3.



Fitted vs. measured binding with ridge



Example

Consider a linear regression model with a single predictor and assume it is known that $\beta_0 = 0$ and $E[X] = 0$. Consider the regularized loss function

$$L_\lambda(\beta_1) = \underbrace{\frac{1}{N}\sum_{i=1}^{N}(\beta_1 X_i - Y_i)^2}_{=L(\beta_1)} + \lambda(\beta_1 - b)^2,$$

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Question

- Write down the formula for $\hat{\beta}_1$, the (unregularized) least squares estimator which minimizes $L(\beta_1)$
- Derive the formula for the regularized estimator

$$\hat{\beta}_1^\lambda = \frac{\hat{\beta}_1}{1 + \lambda/\hat{\sigma}_X^2} + \frac{\lambda/\hat{\sigma}_X^2}{1 + \lambda/\hat{\sigma}_X^2}\, b$$

Least square with uncorrelated features

Consider a linear regression model with K features and assume it is known that $\beta_0 = 0$:

$$Y = \beta_1 \phi_1(X) + \cdots + \beta_K \phi_K(X) + \epsilon.$$

Suppose the features are uncorrelated and mean zero in the data:

$$\frac{1}{N} \sum_{i=1}^{N} \phi_j(X_i)\phi_k(X_i) = 0, \quad j \neq k, \qquad \frac{1}{N} \sum_{i=1}^{N} \phi_k(X_i) = 0.$$

Consider the least-squares loss function

$$L(\beta_1, \ldots, \beta_K) = \frac{1}{N} \sum_{i=1}^{N} \Big(\sum_{k=1}^{K} \beta_k \phi_k(X_i) - Y_i \Big)^2.$$

Question

What are the least-squares estimators $\hat{\beta}_k$ that minimize L.

Solution: least squares with uncorrelated features

Let β'_k be the slope when Y is regressed on $\phi_k(X)$ alone. By the omitted variable bias formula,

$$\beta'_k = \beta_k + \sum_{j \neq k} \beta_j \beta_{k,j}, \qquad \beta_{k,j} = \frac{\operatorname{cov}\big(\phi_k(X), \phi_j(X)\big)}{\operatorname{var}\big(\phi_k(X)\big)}.$$

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With the covariances computed from the data, and the features mean zero,

$$\hat{\beta}_k = \frac{\frac{1}{N}\sum_{i=1}^N \phi_k(X_i) Y_i}{\frac{1}{N}\sum_{i=1}^N \phi_k(X_i)^2} = \frac{\sum_{i=1}^N \phi_k(X_i) Y_i}{\sum_{i=1}^N \phi_k(X_i)^2}.$$

Regularized least squares with uncorrelated features

Same model, with the features uncorrelated and mean zero in the data. Ridge regression minimizes

$$L_\lambda(\beta_1, \ldots, \beta_K) = \frac{1}{N} \sum_{i=1}^{N} \Big(\sum_{k=1}^{K} \beta_k \phi_k(X_i) - Y_i \Big)^2 + \lambda \sum_{k=1}^{K} \beta_k^2.$$

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Because the features are uncorrelated, each β_k is found on its own:

$$\hat{\beta}_k^\lambda = \frac{\frac{1}{N} \sum_{i=1}^{N} \phi_k(X_i)^2}{\frac{1}{N} \sum_{i=1}^{N} \phi_k(X_i)^2 + \lambda} \hat{\beta}_k,$$

where $\hat{\beta}_k$ is the least-squares estimator.

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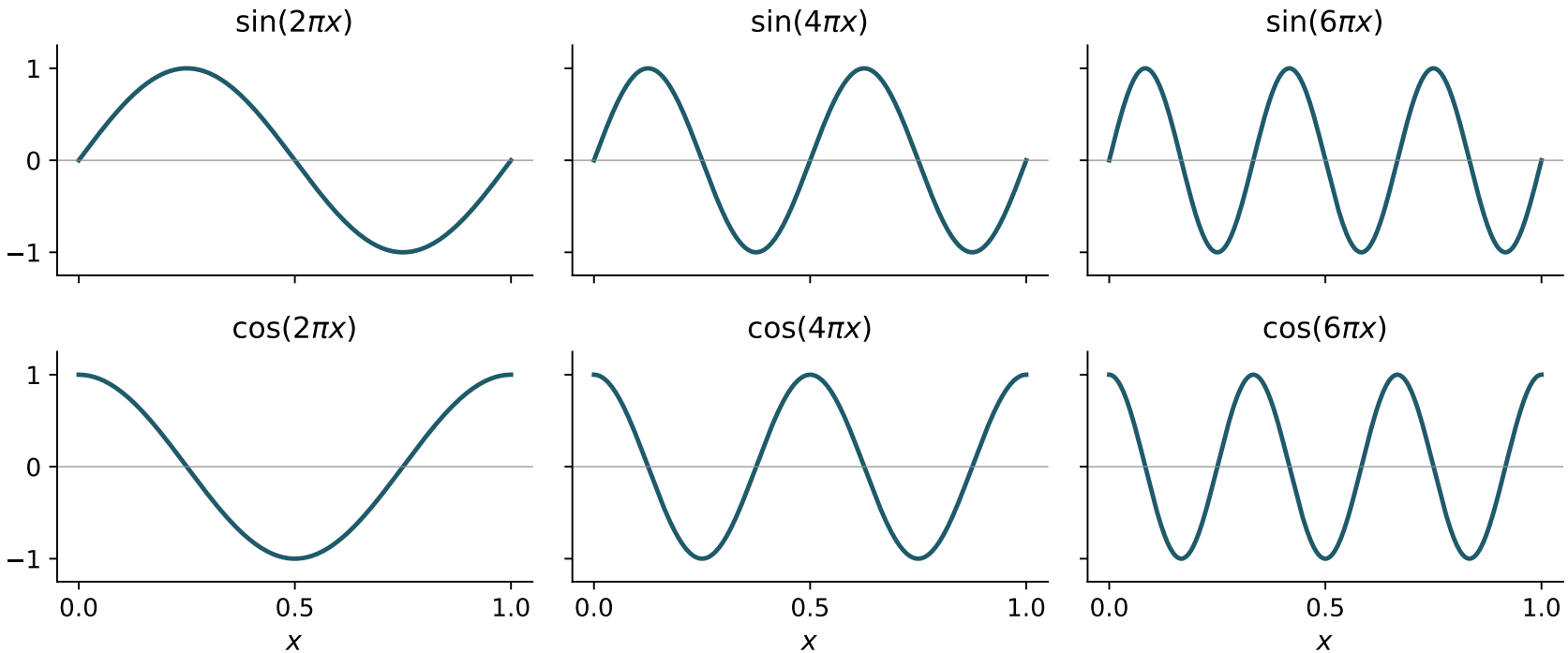
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This is analogous to the single-predictor result $\hat{\beta}_1^\lambda = \dfrac{\hat{\sigma}_X^2}{\hat{\sigma}_X^2 + \lambda} \hat{\beta}_1$, applied to each feature separately.

Sines and cosines as features

For $k = 1, 2, 3, \ldots$, the functions $\sin(2\pi kx)$ and $\cos(2\pi kx)$ are waves with k cycles on $[0, 1)$.



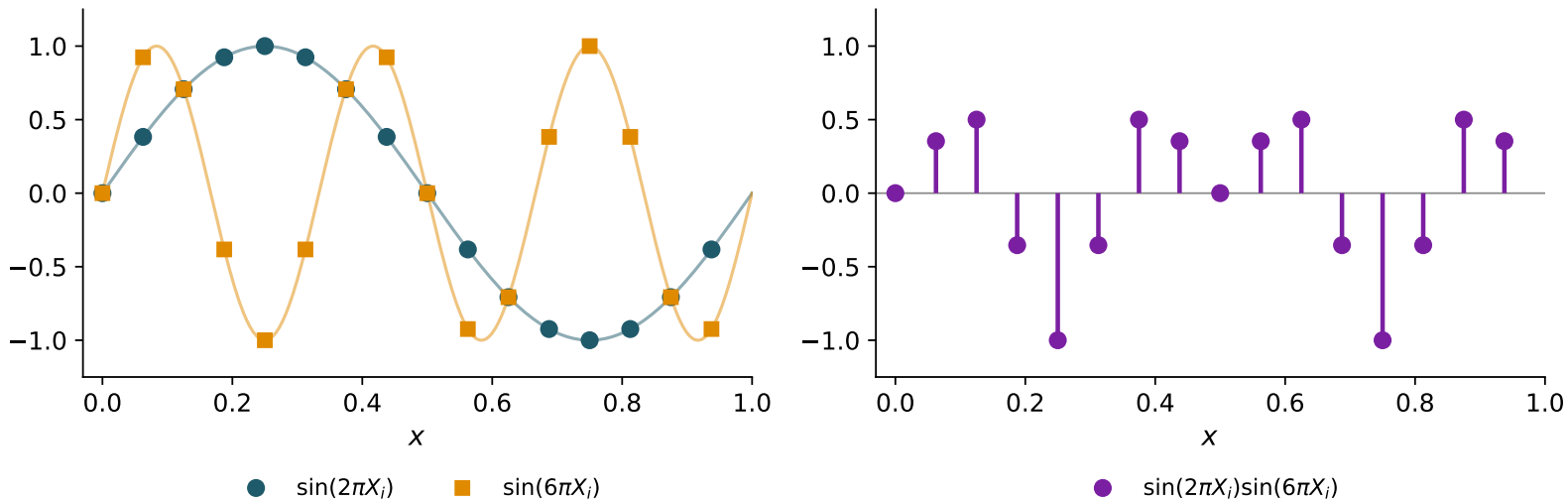
Sines and cosines as features

Theorem

Let $X_i = (i-1)/N$ for $i = 1, \ldots, N$, and let j, k be integers with $0 < j, k < N/2$. Then

$$\sum_{i=1}^{N} \sin(2\pi j X_i) \sin(2\pi k X_i) = \sum_{i=1}^{N} \cos(2\pi j X_i) \cos(2\pi k X_i) = 0 \quad (j \neq k),$$

$$\sum_{i=1}^{N} \sin(2\pi j X_i) \cos(2\pi k X_i) = 0 \quad (\text{all } j, k), \qquad \sum_{i=1}^{N} \sin^2(2\pi k X_i) = \sum_{i=1}^{N} \cos^2(2\pi k X_i) = N/2.$$



Choosing the features: sines and cosines

Sins and cosines are uncorrelated when the X points are equally spaced, therefore we select the features

$$\sin(2\pi kx),\ \cos(2\pi kx), \quad k = 1, \ldots, K$$

which gives the Fourier model

$$Y = \beta_0 + \sum_{k=1}^{K} \Big[\beta_k \sin(2\pi kX) + \alpha_k \cos(2\pi kX)\Big] + \epsilon.$$

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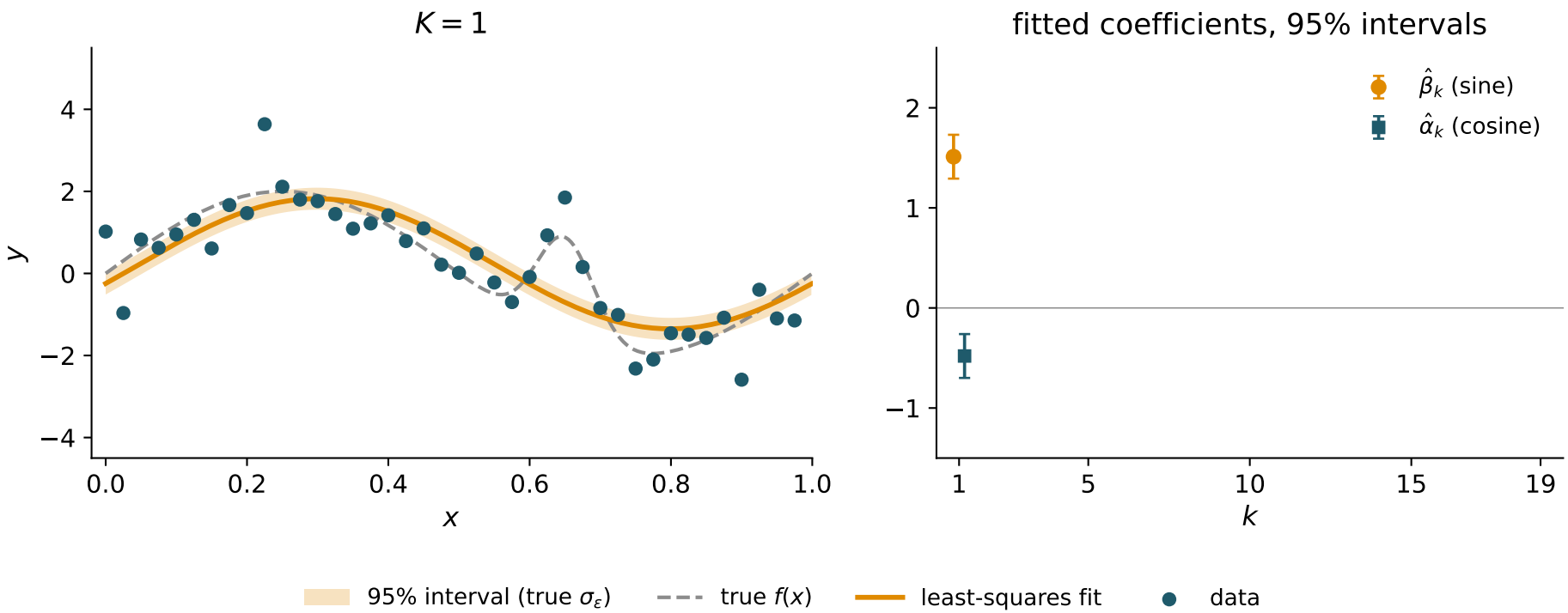
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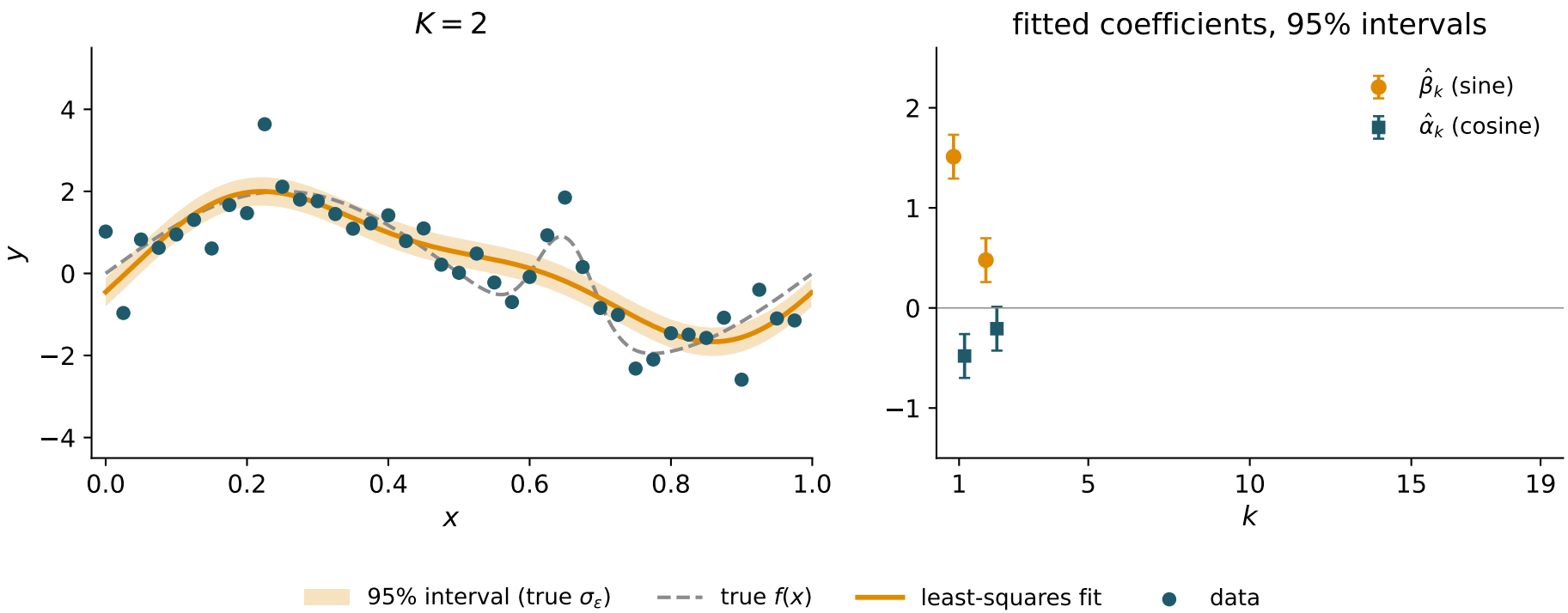
Using that the features are uncorrelated and $\sum_i \sin^2(2\pi kX_i) = \sum_i \cos^2(2\pi kX_i) = N/2$,

$$\hat{\beta}_k = \frac{2}{N}\sum_{i=1}^{N} \sin(2\pi kX_i)\, Y_i, \qquad \hat{\alpha}_k = \frac{2}{N}\sum_{i=1}^{N} \cos(2\pi kX_i)\, Y_i, \qquad \hat{\beta}_0 = \overline{Y}.$$

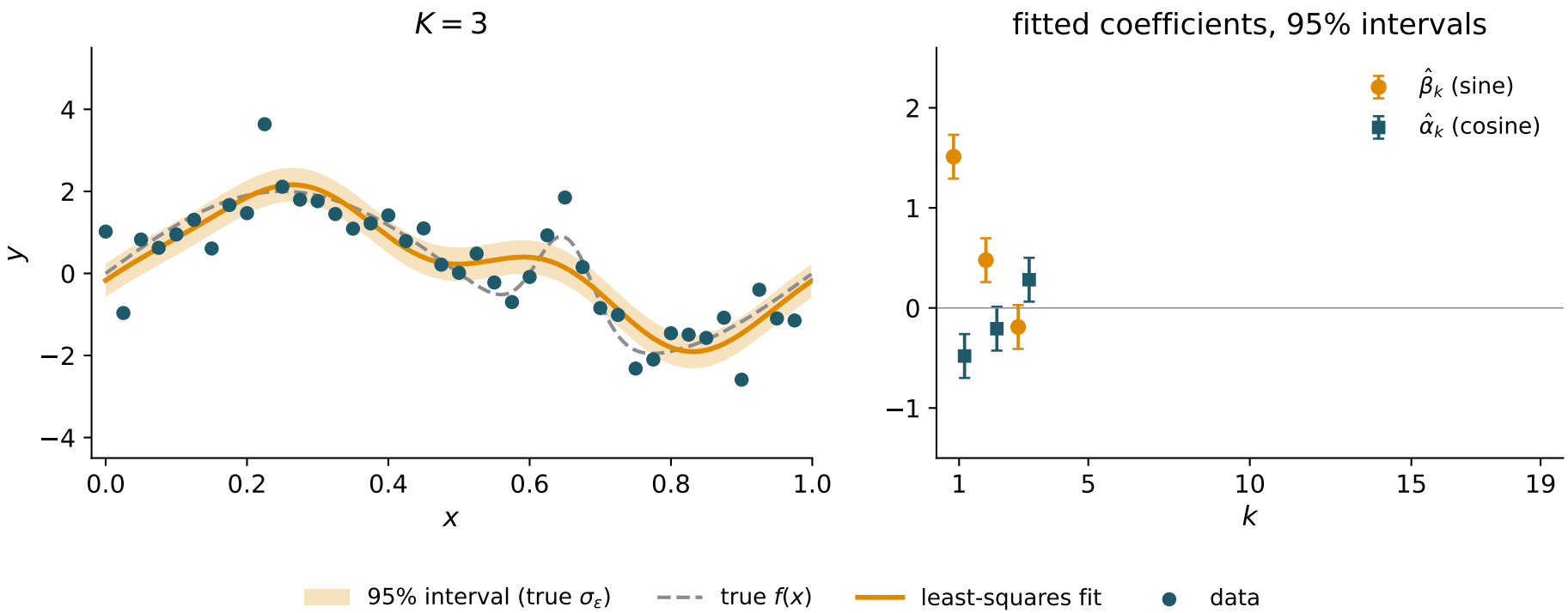
Example: dependence on K



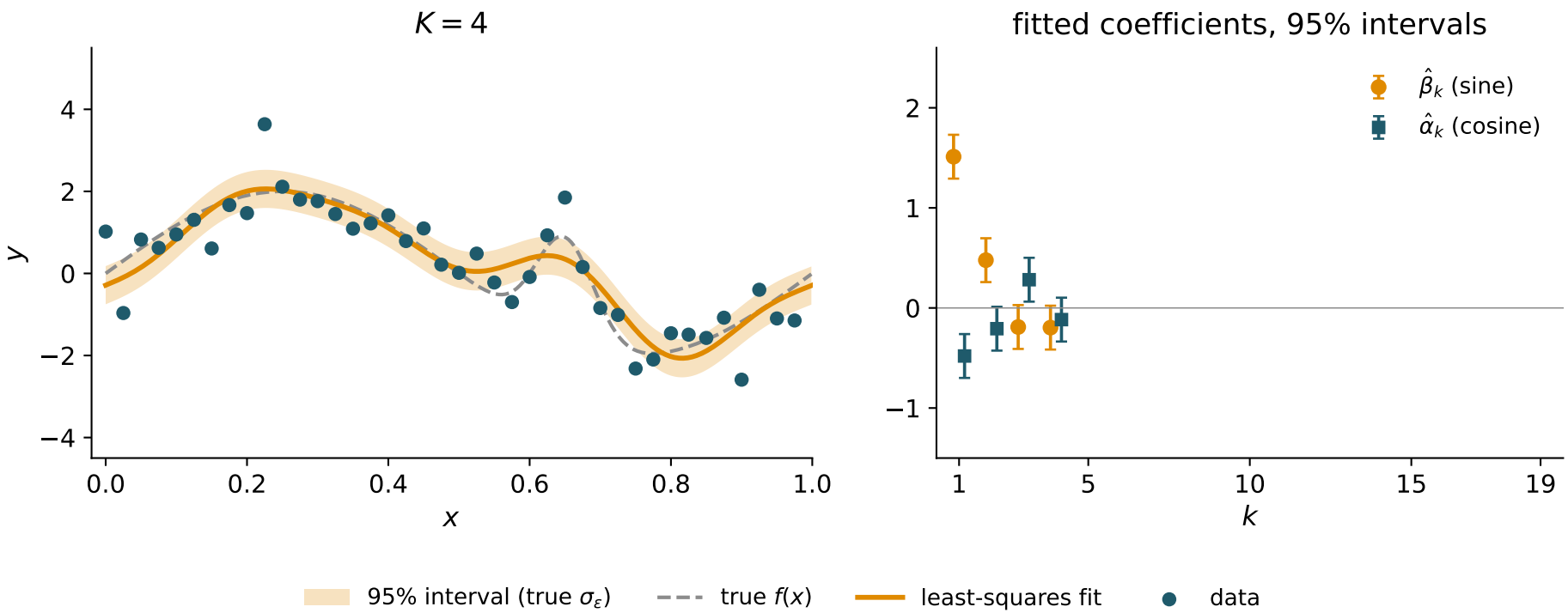
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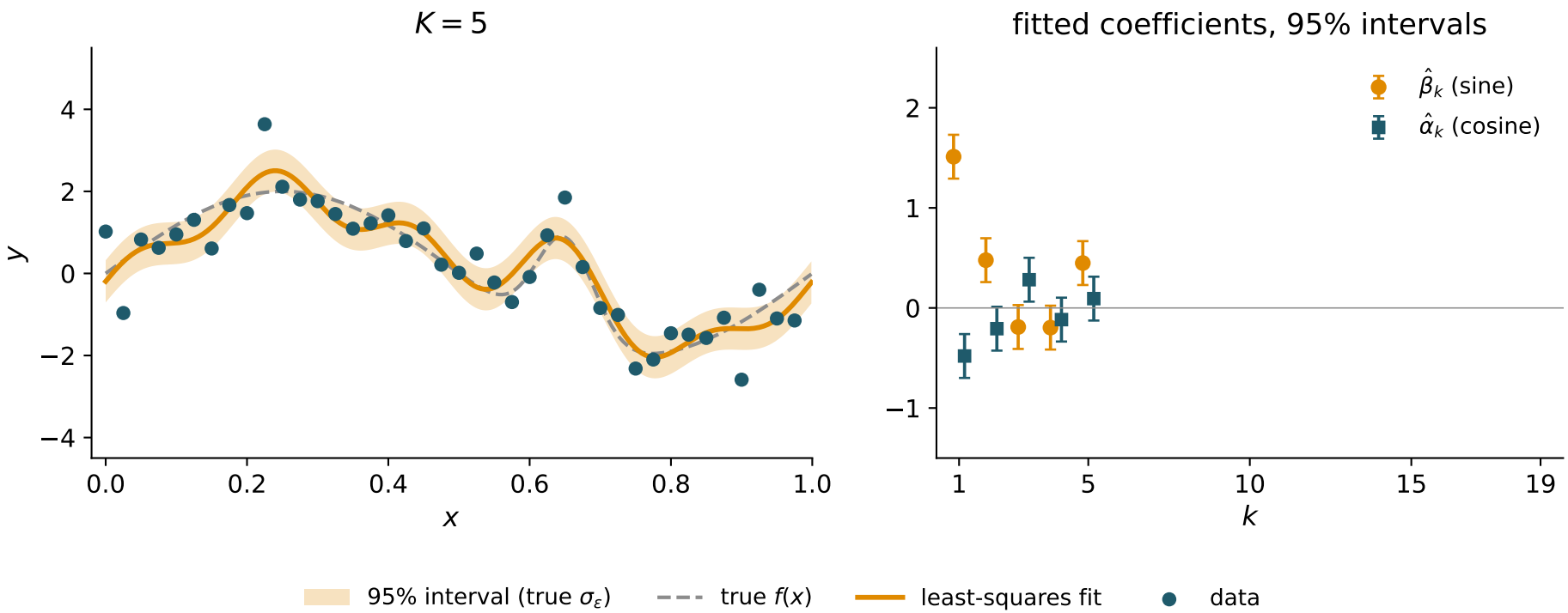
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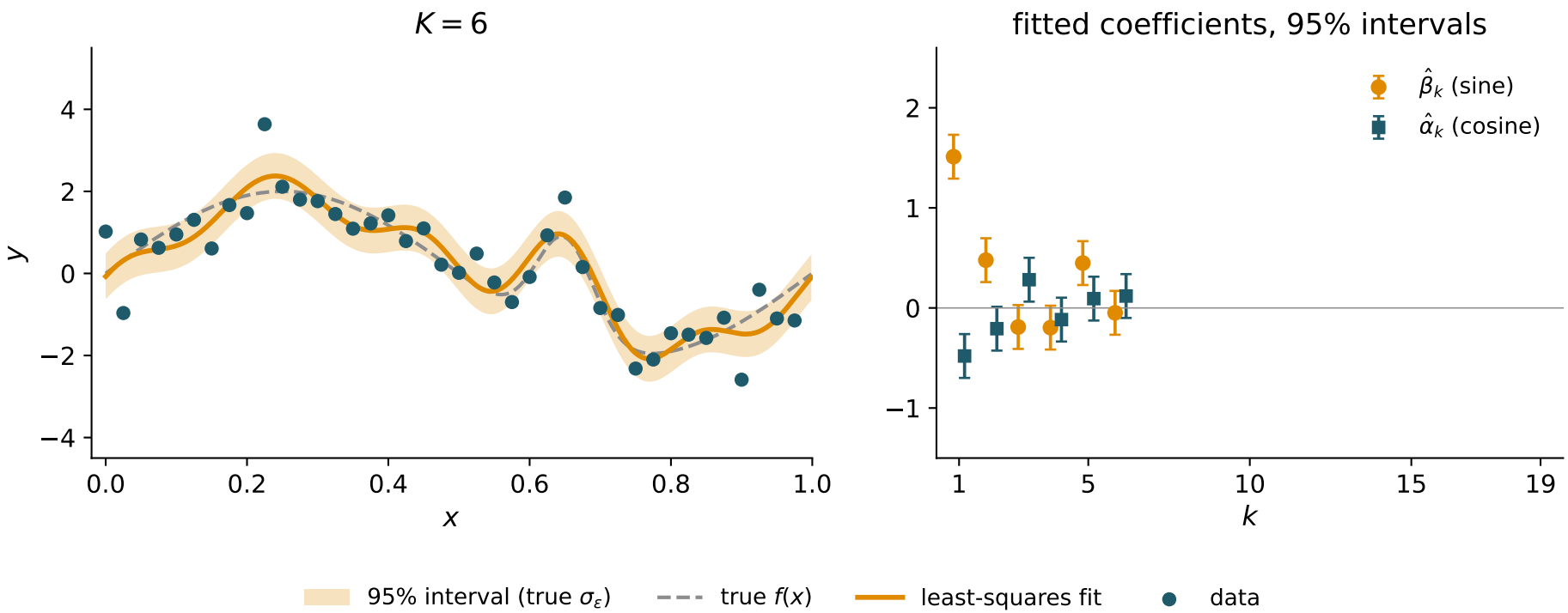
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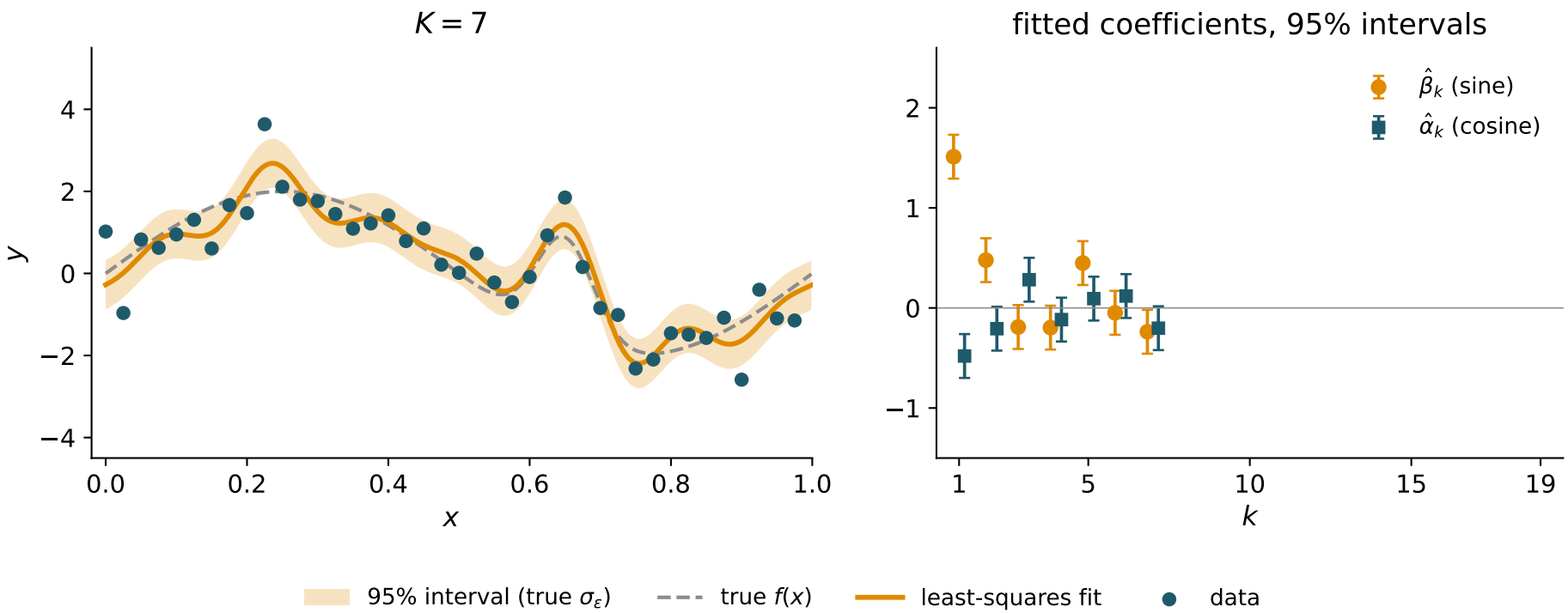
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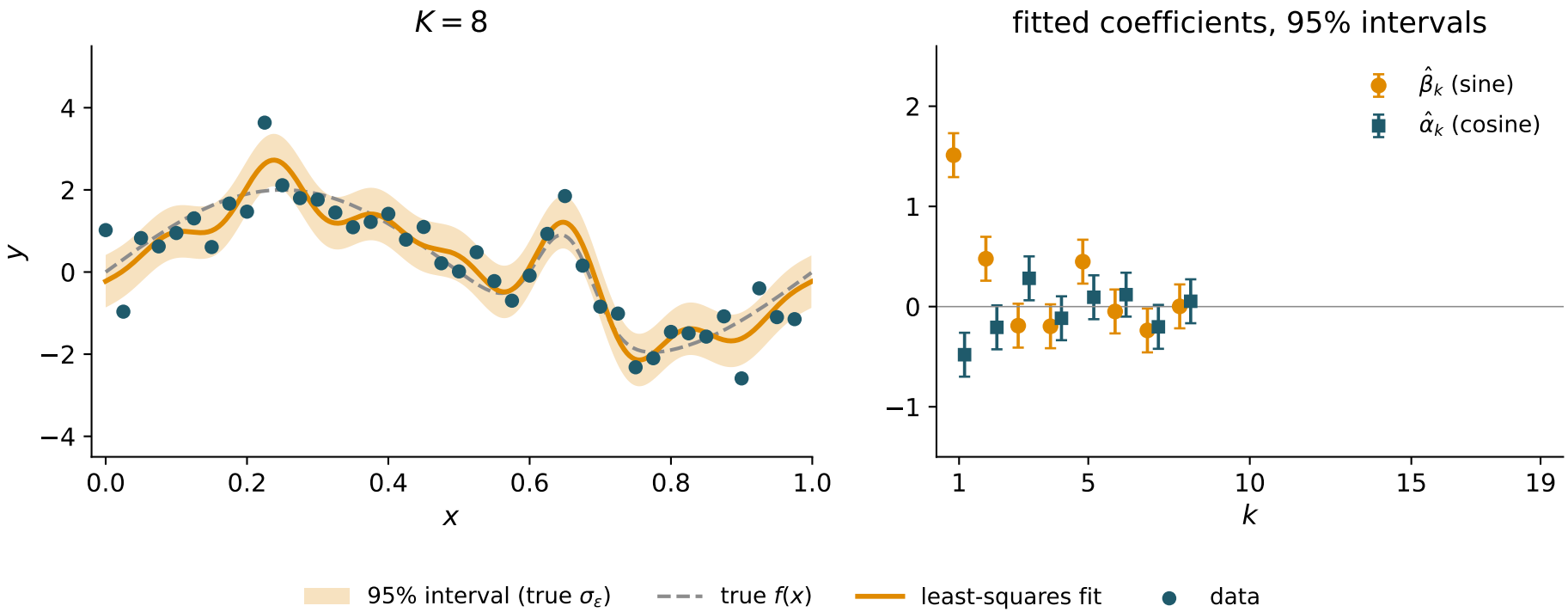
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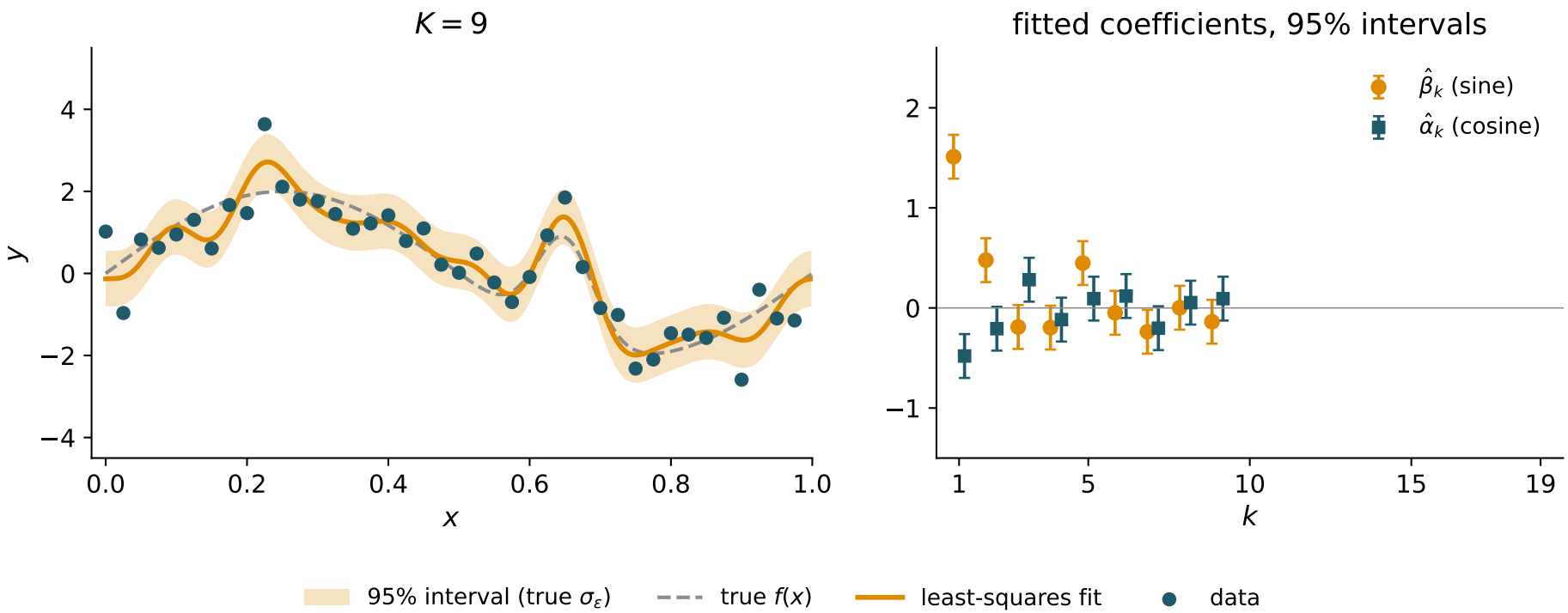
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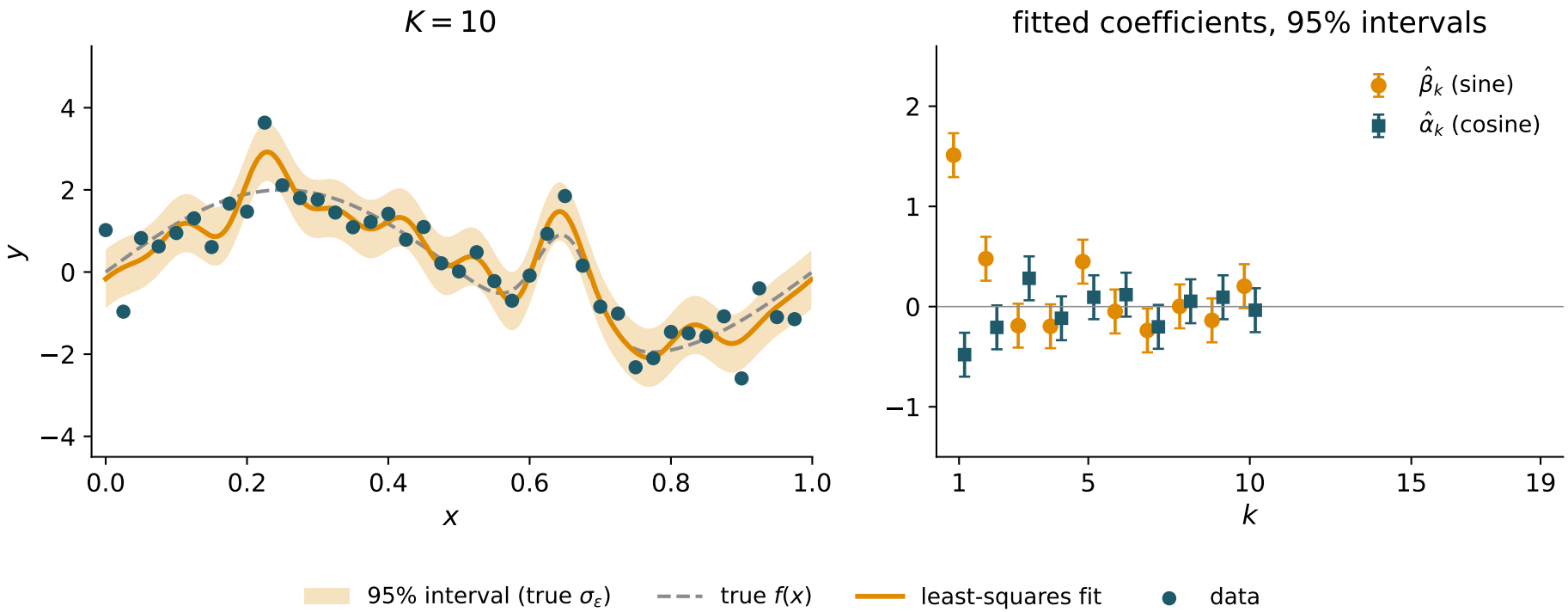
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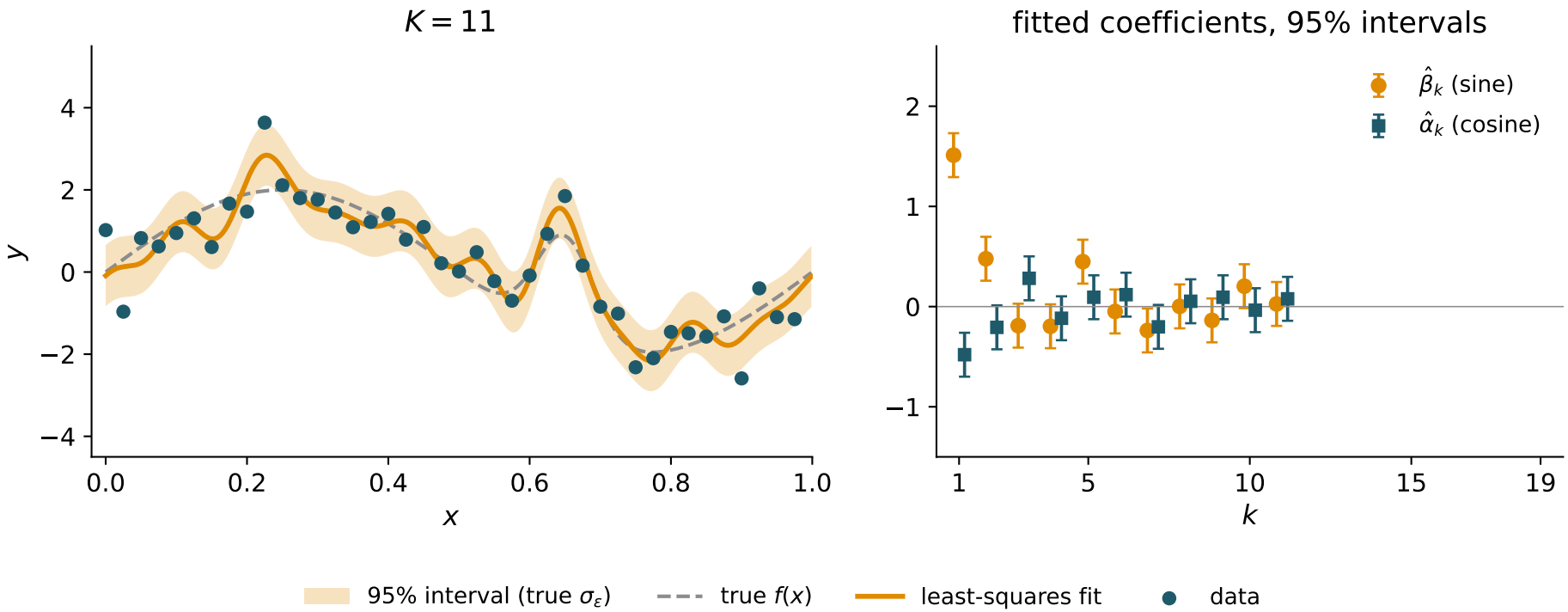
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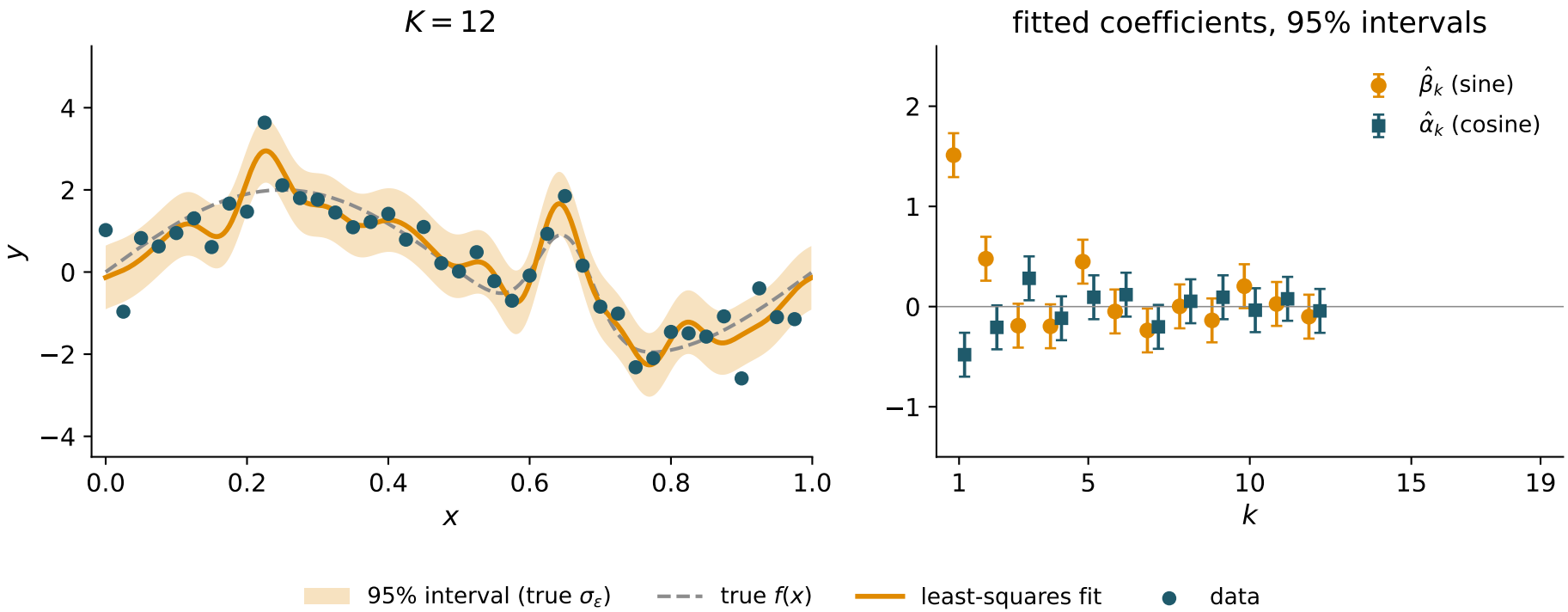
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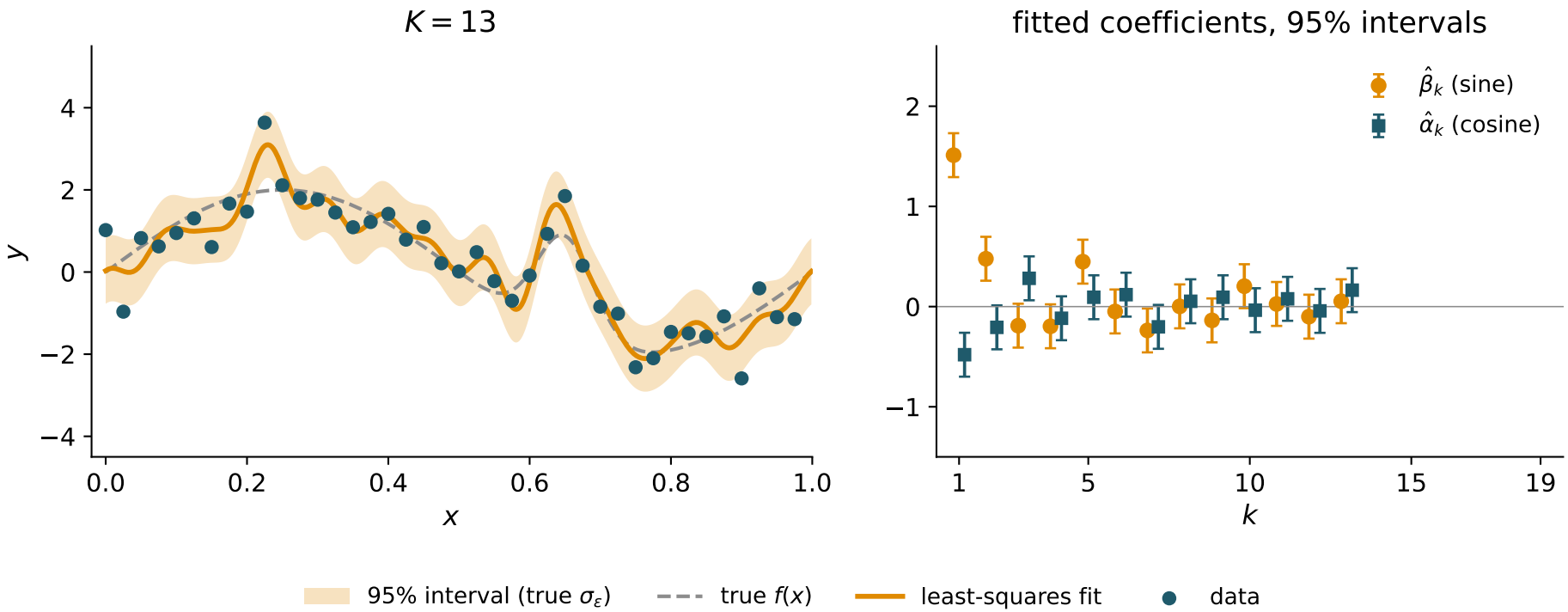
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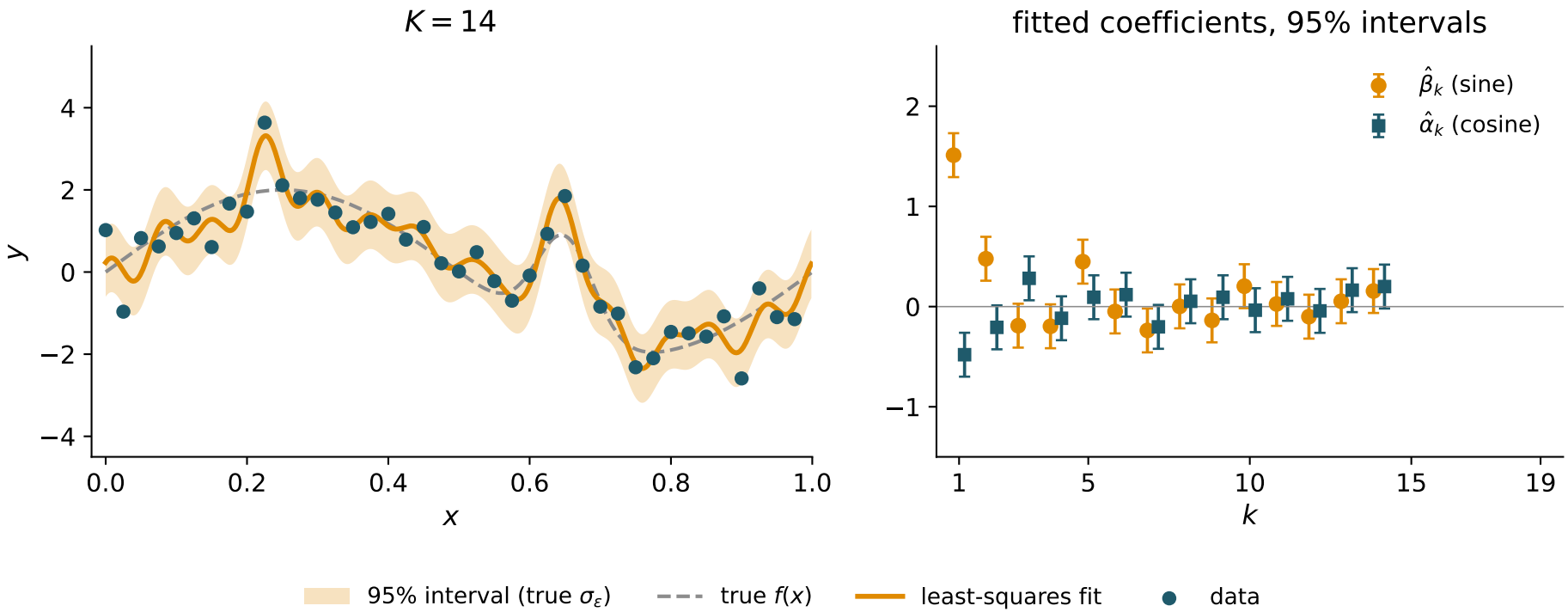
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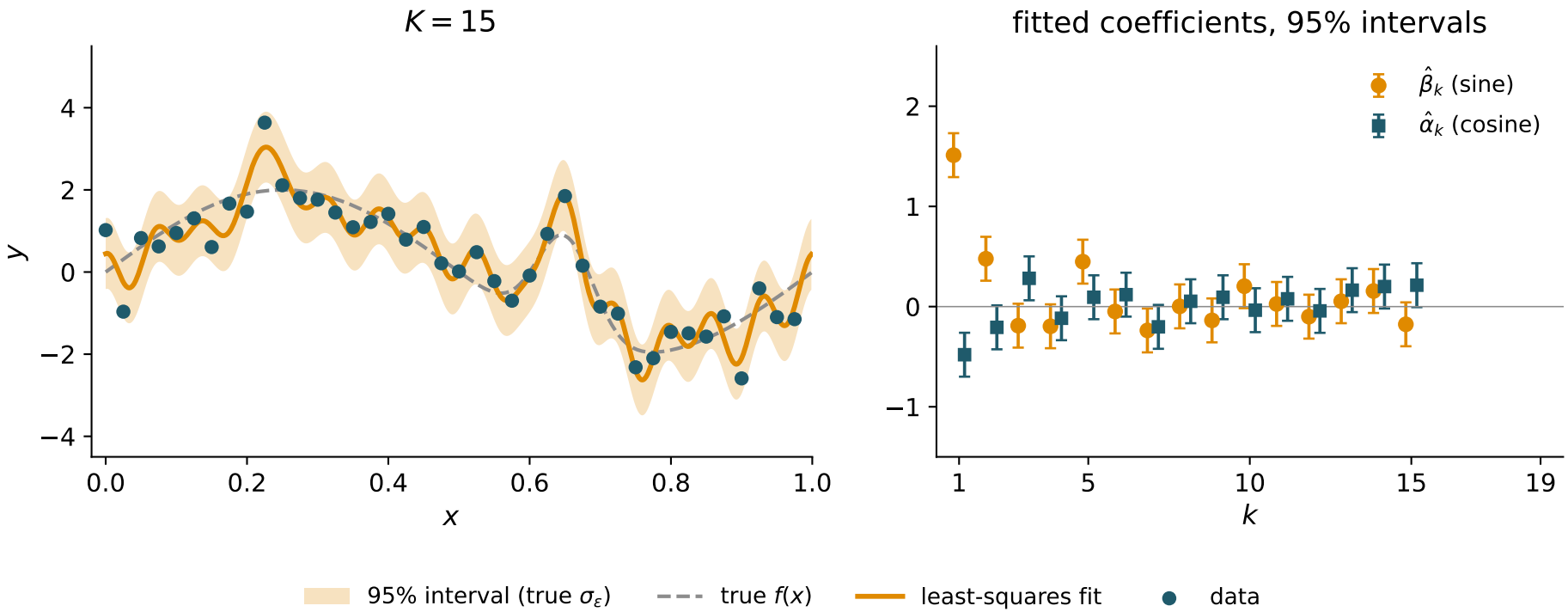
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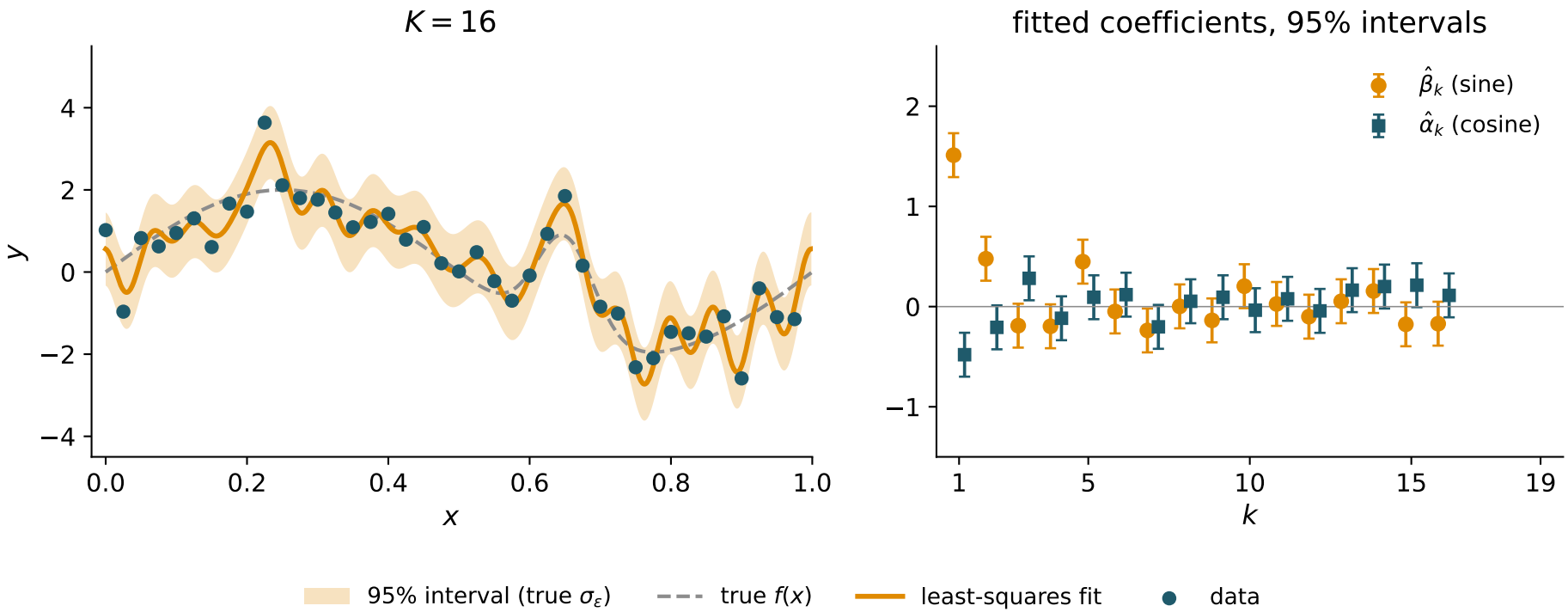
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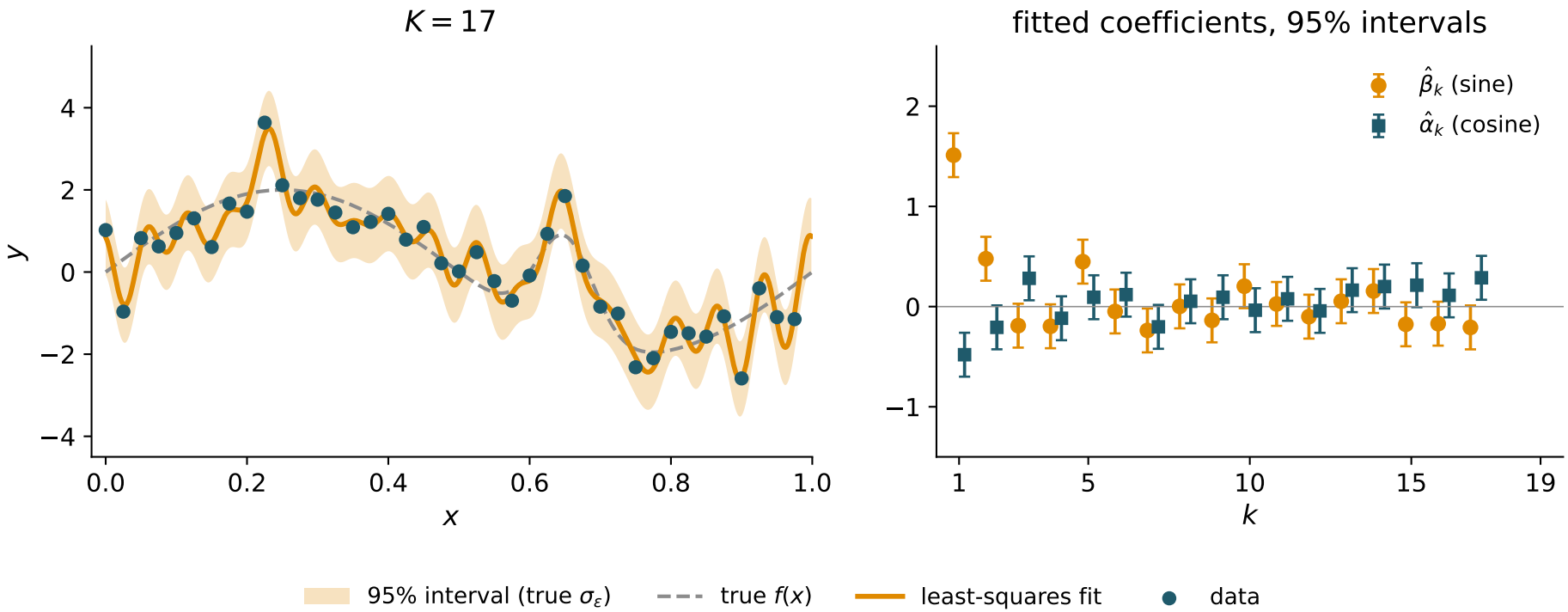
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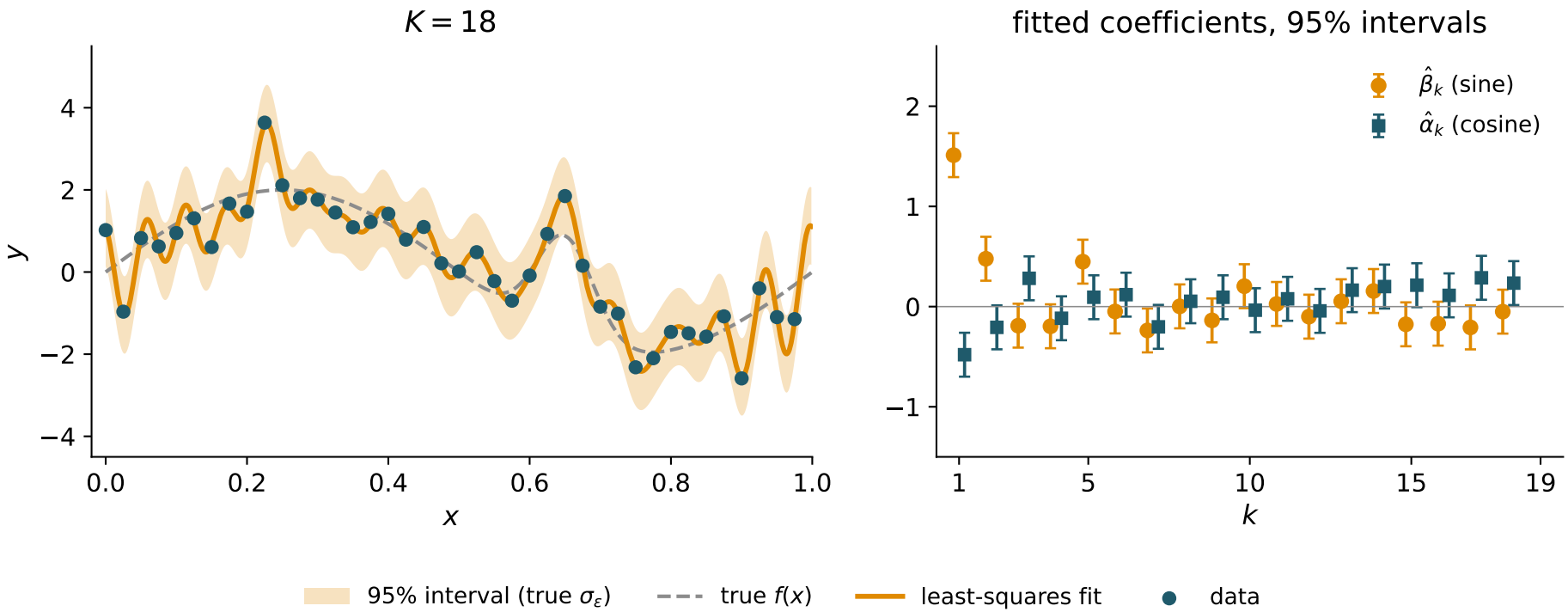
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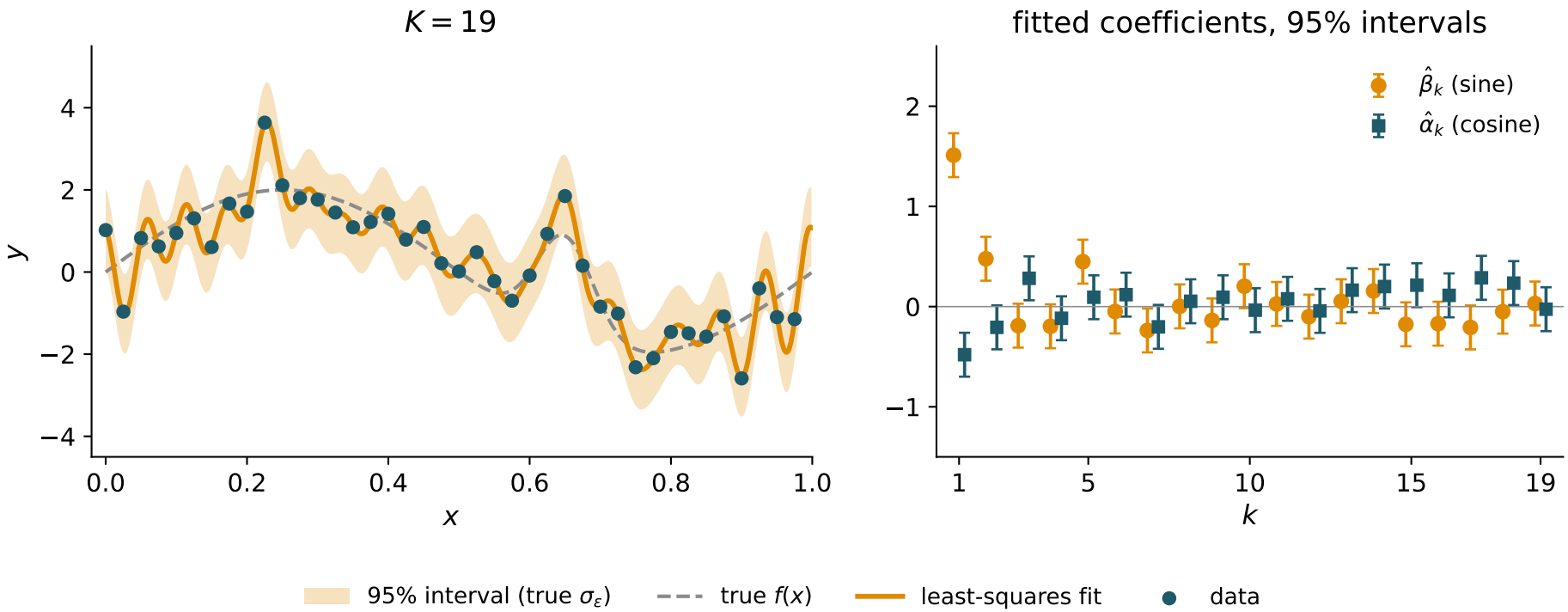
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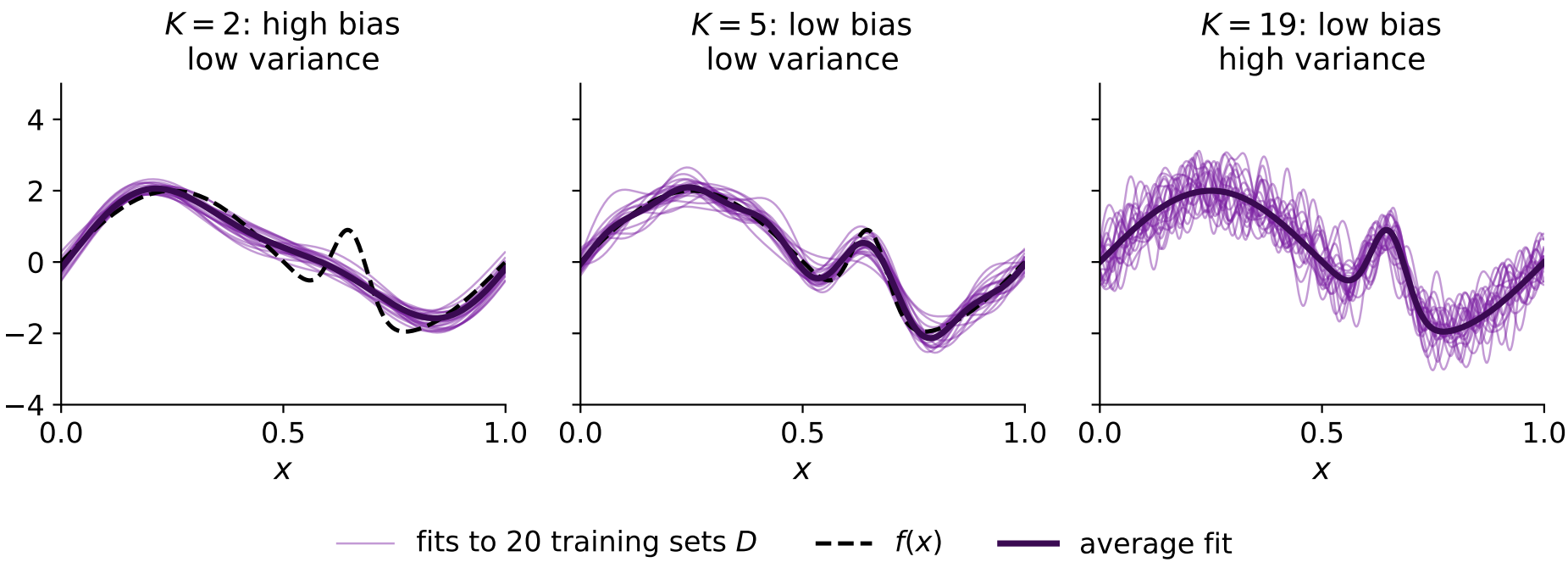
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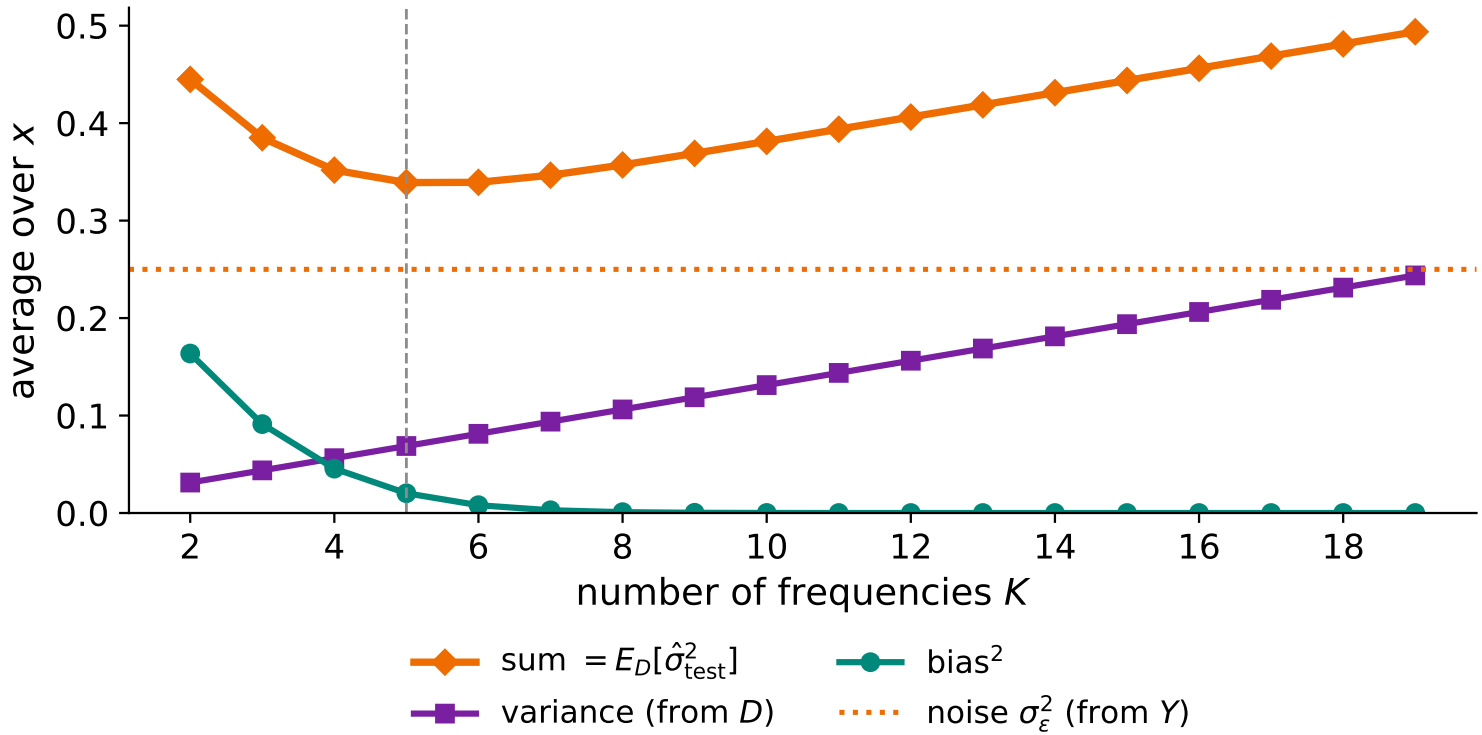
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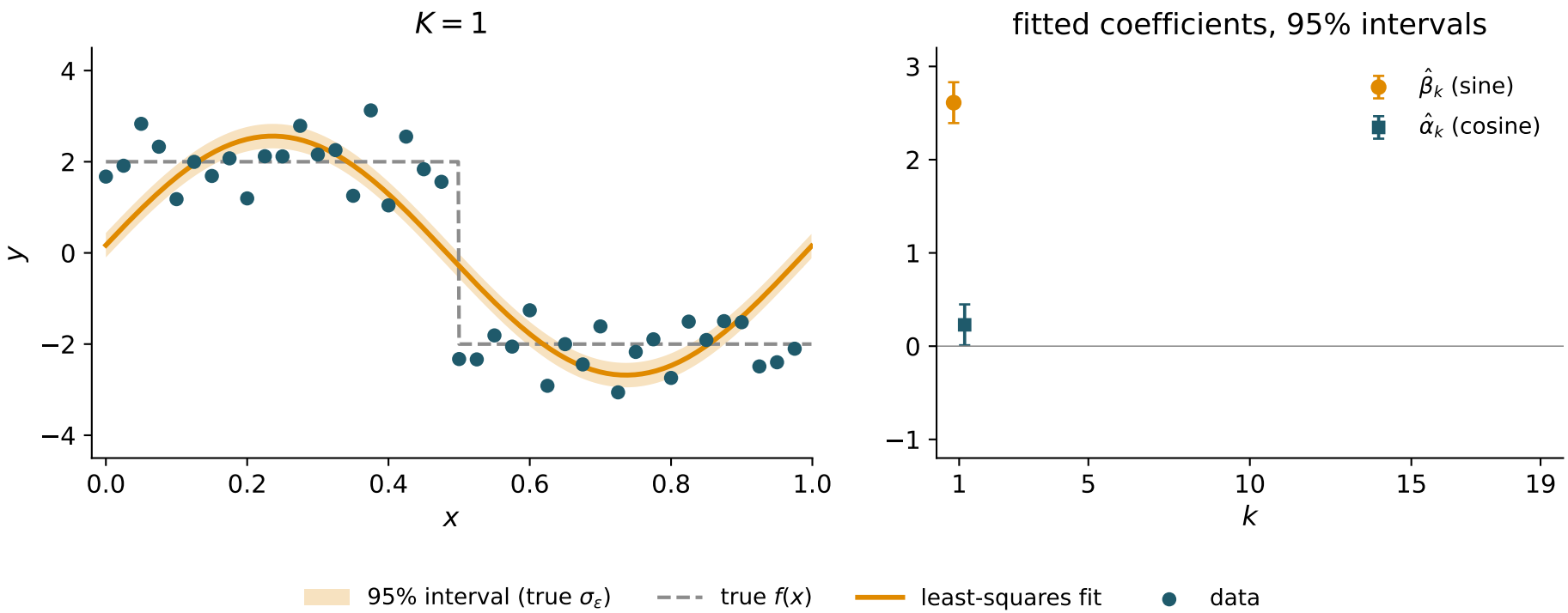
Fits to many training sets



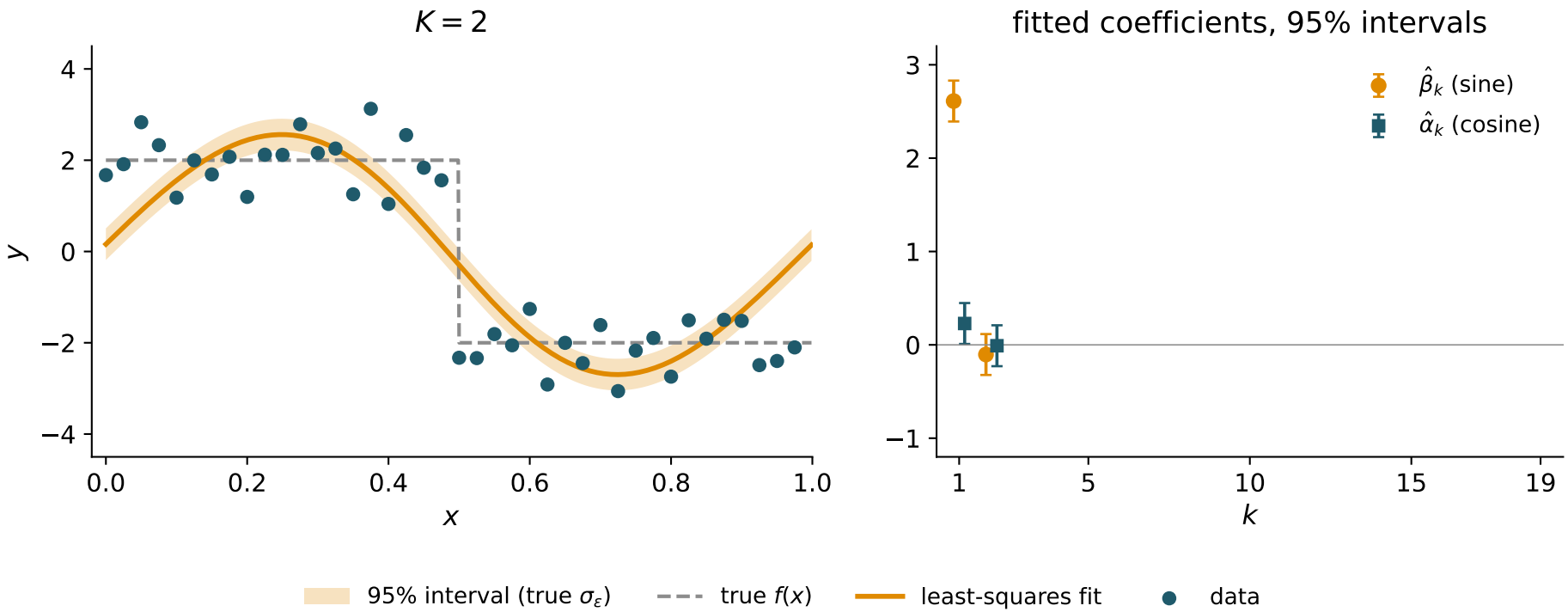
Bias and variance vs. K



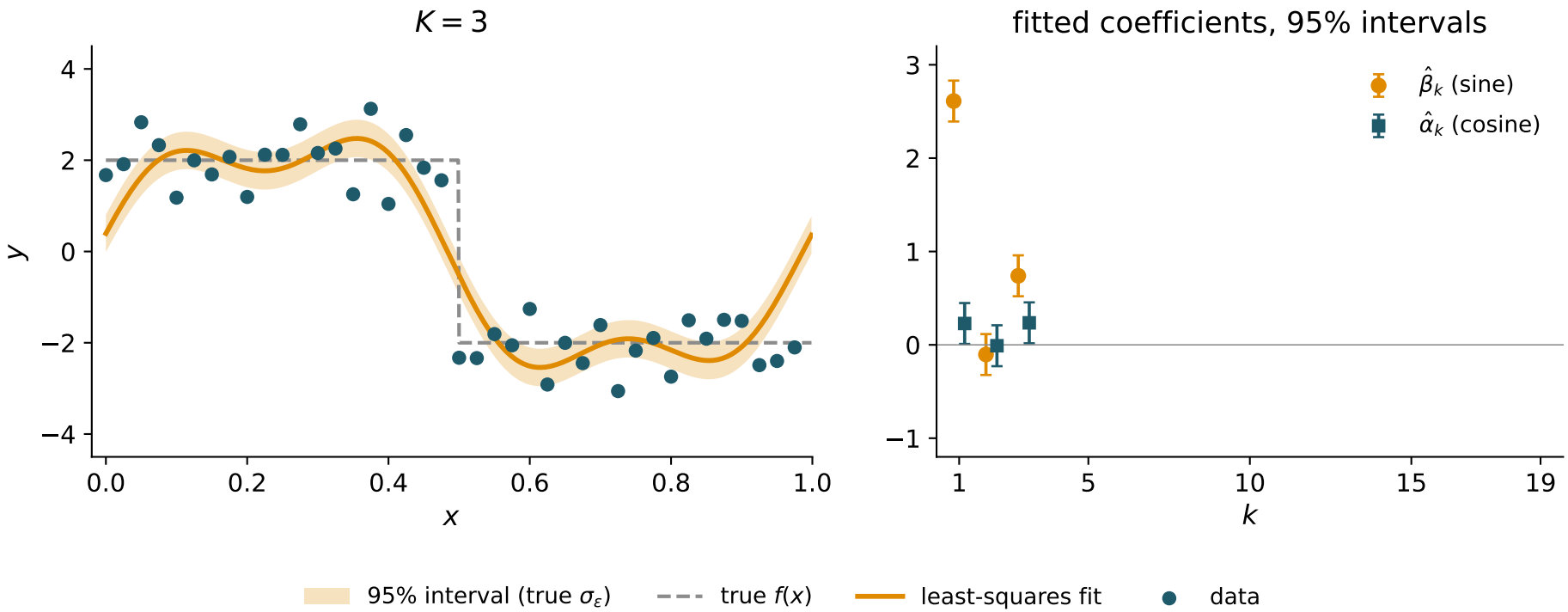
Example: a square wave



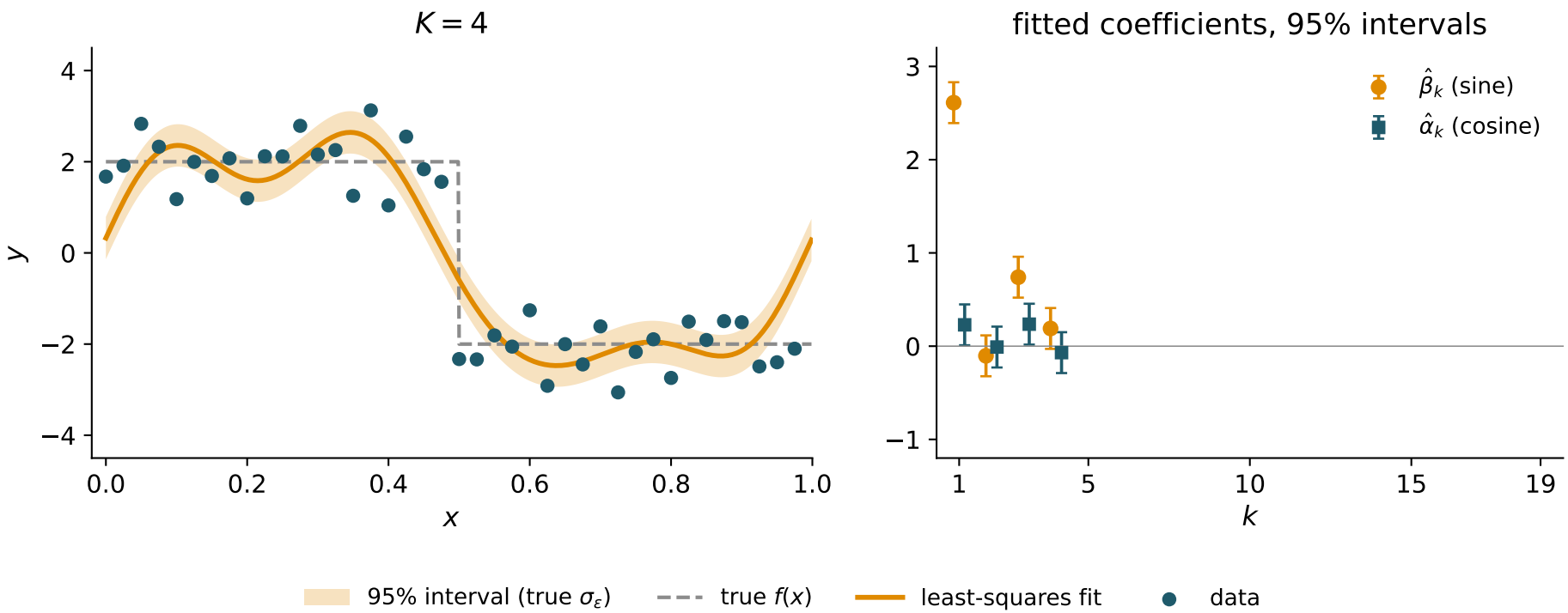
Example: a square wave



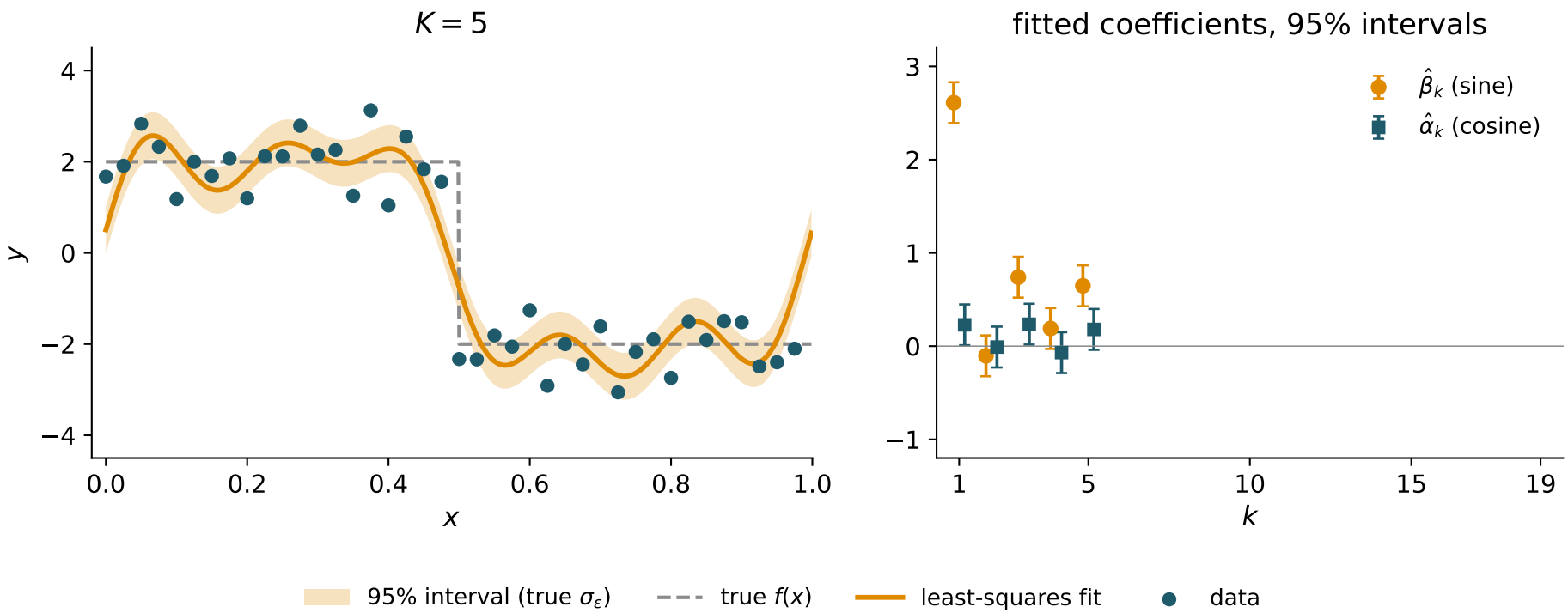
Example: a square wave



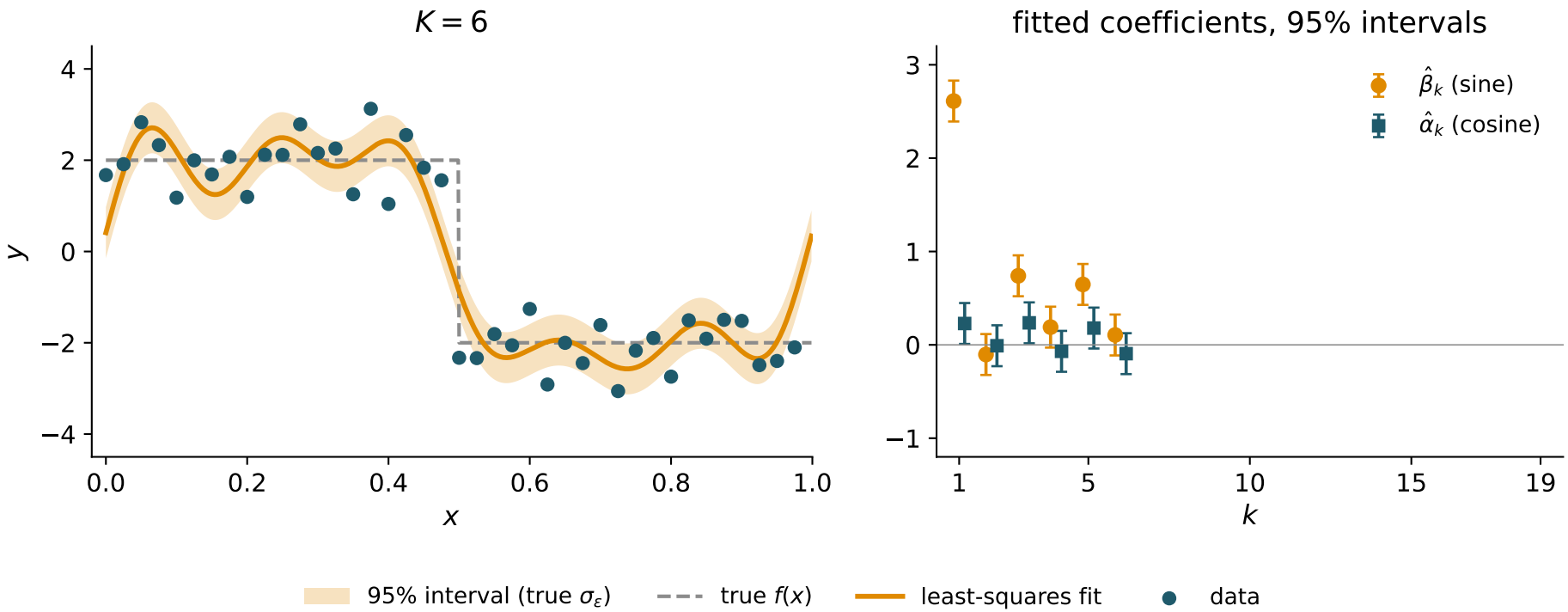
Example: a square wave



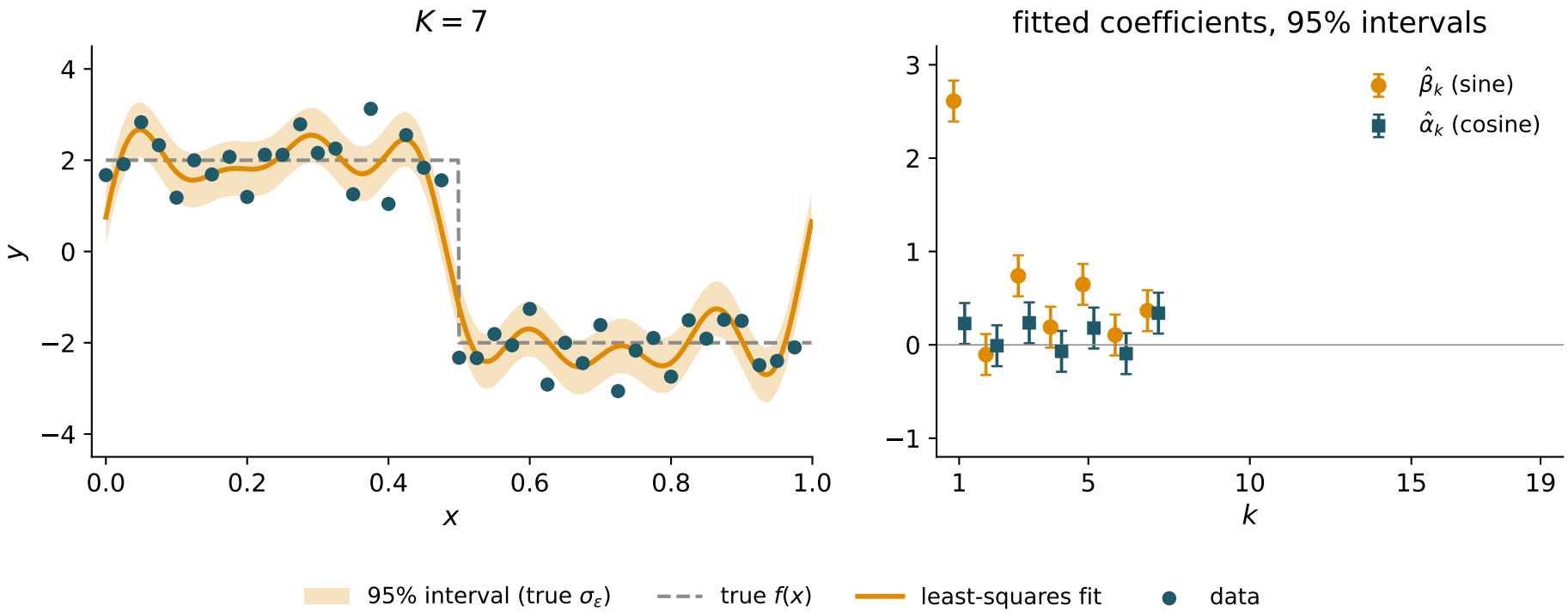
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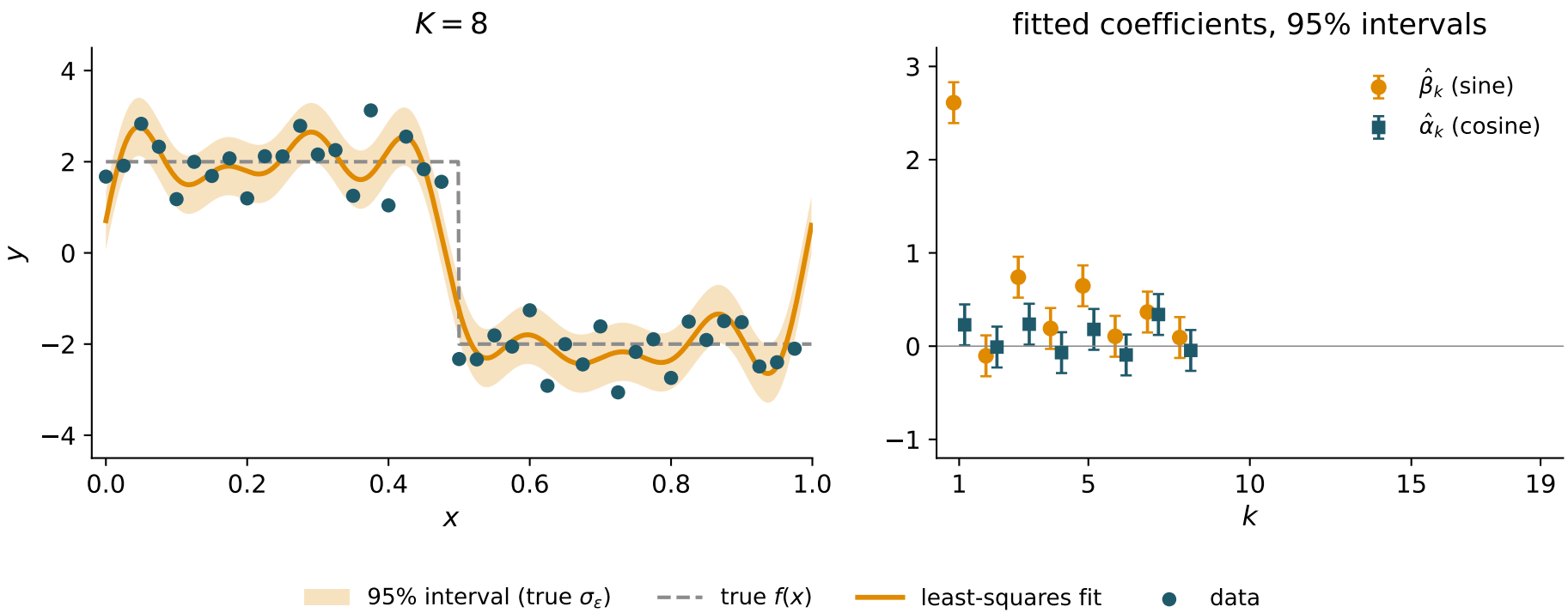
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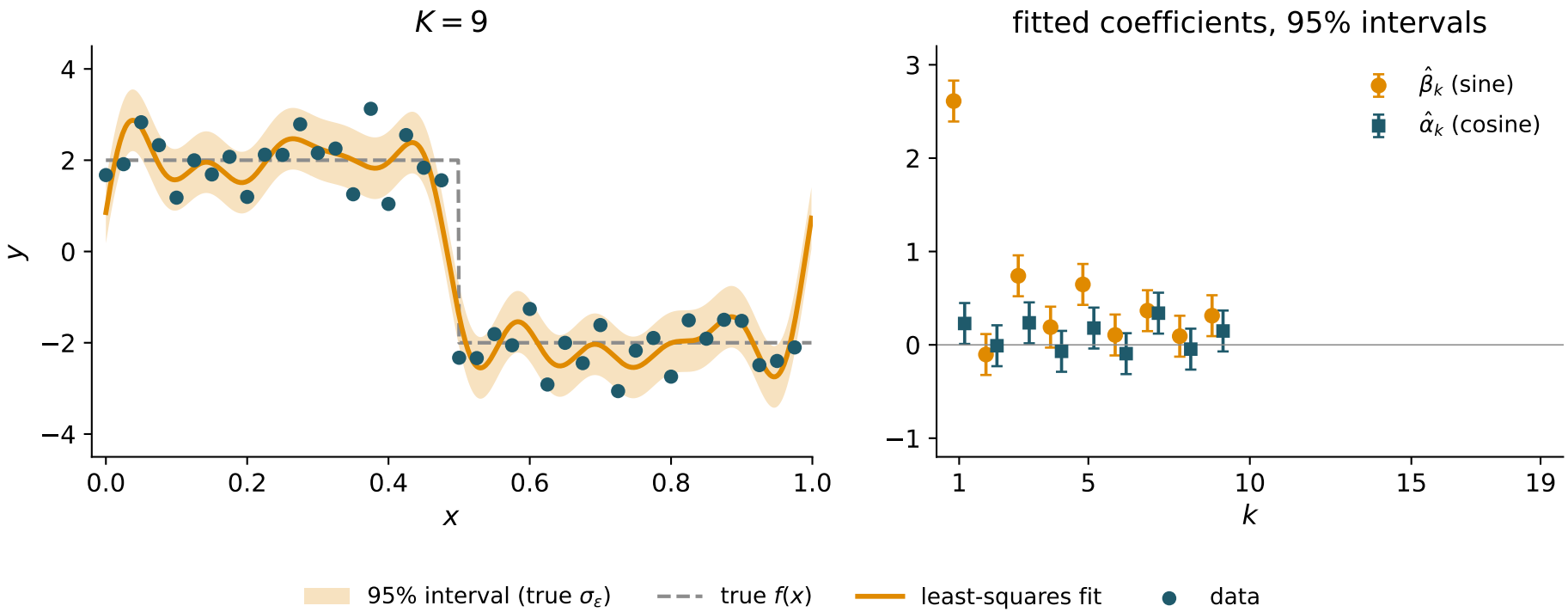
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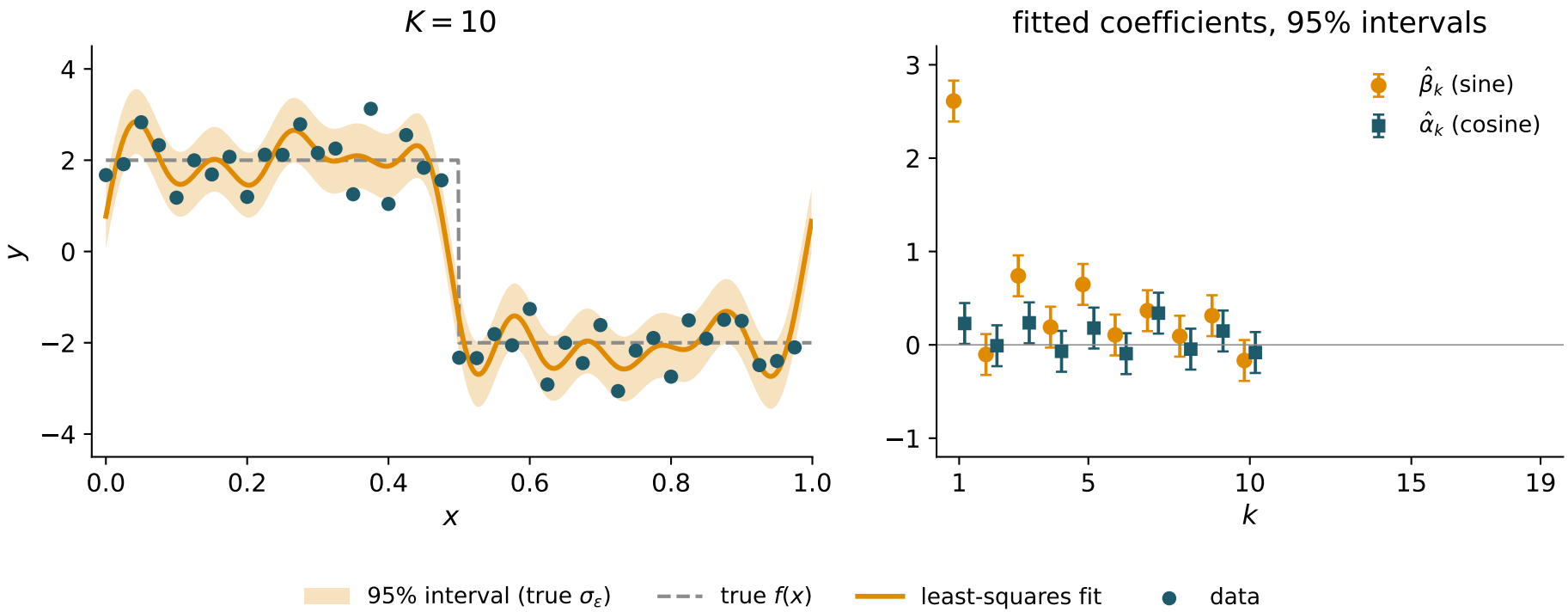
Example: a square wave



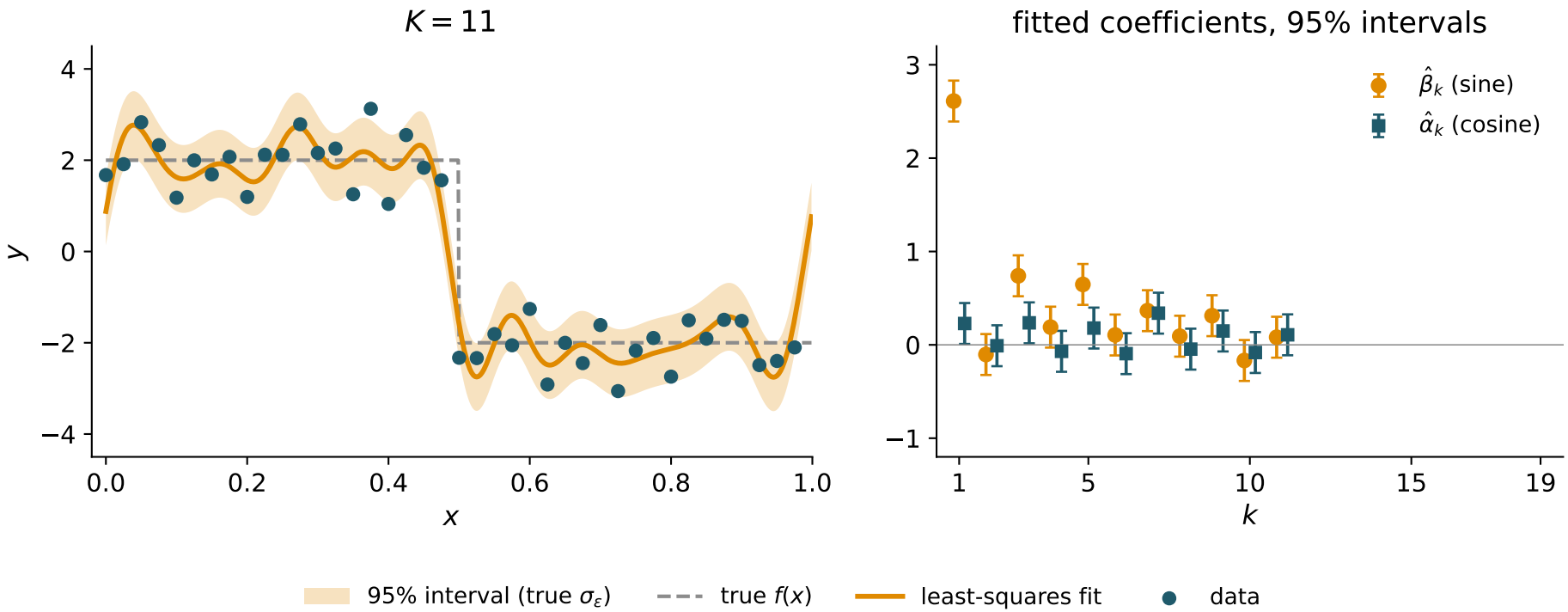
Example: a square wave



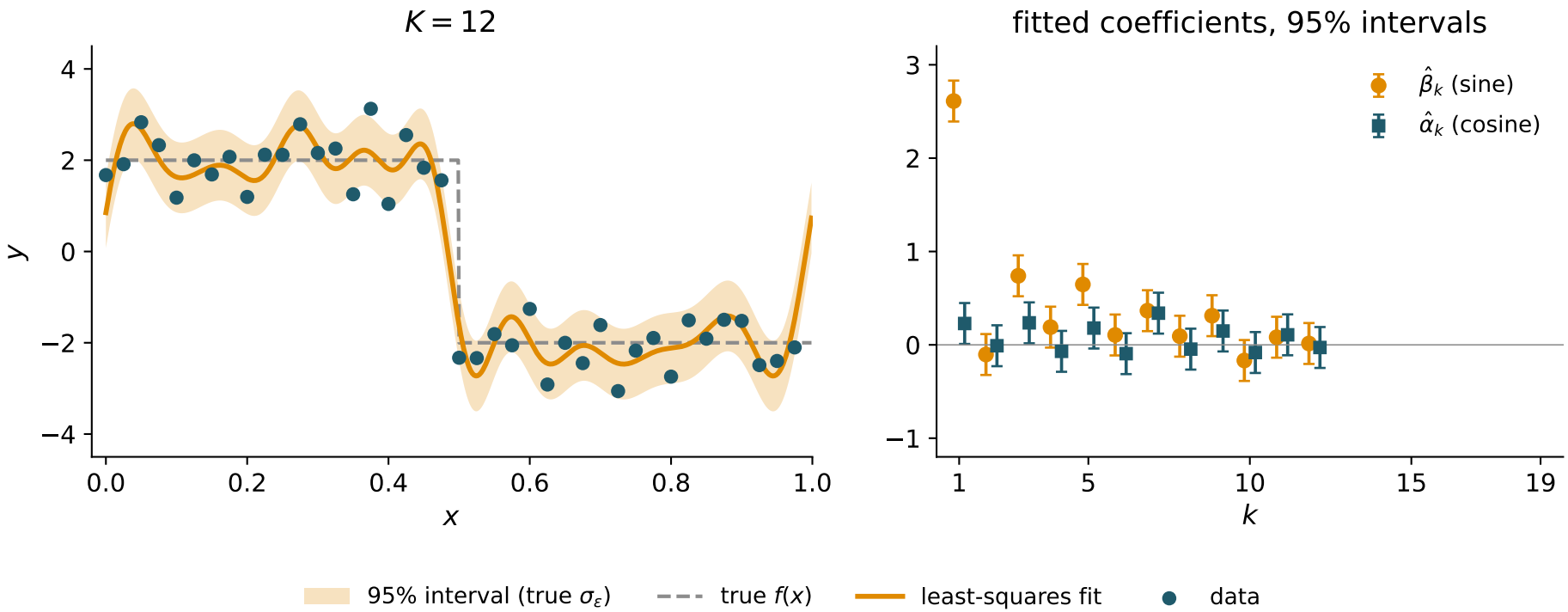
Example: a square wave



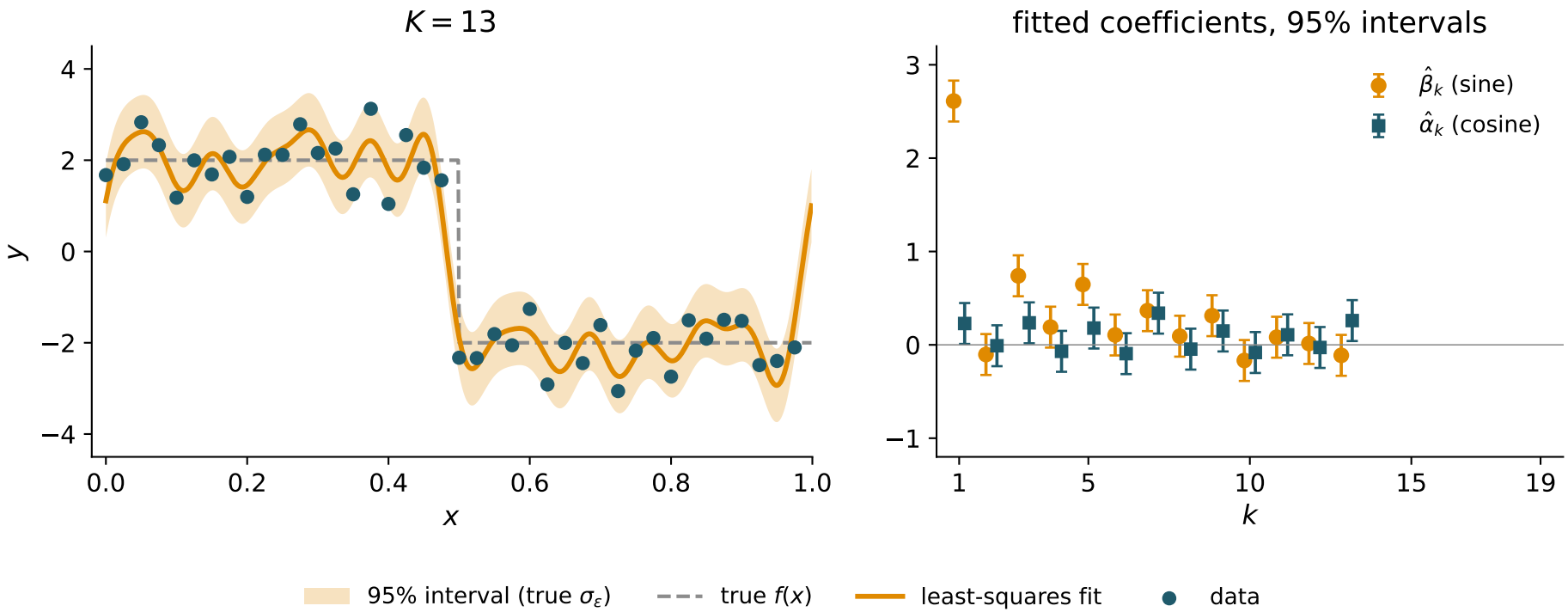
Example: a square wave



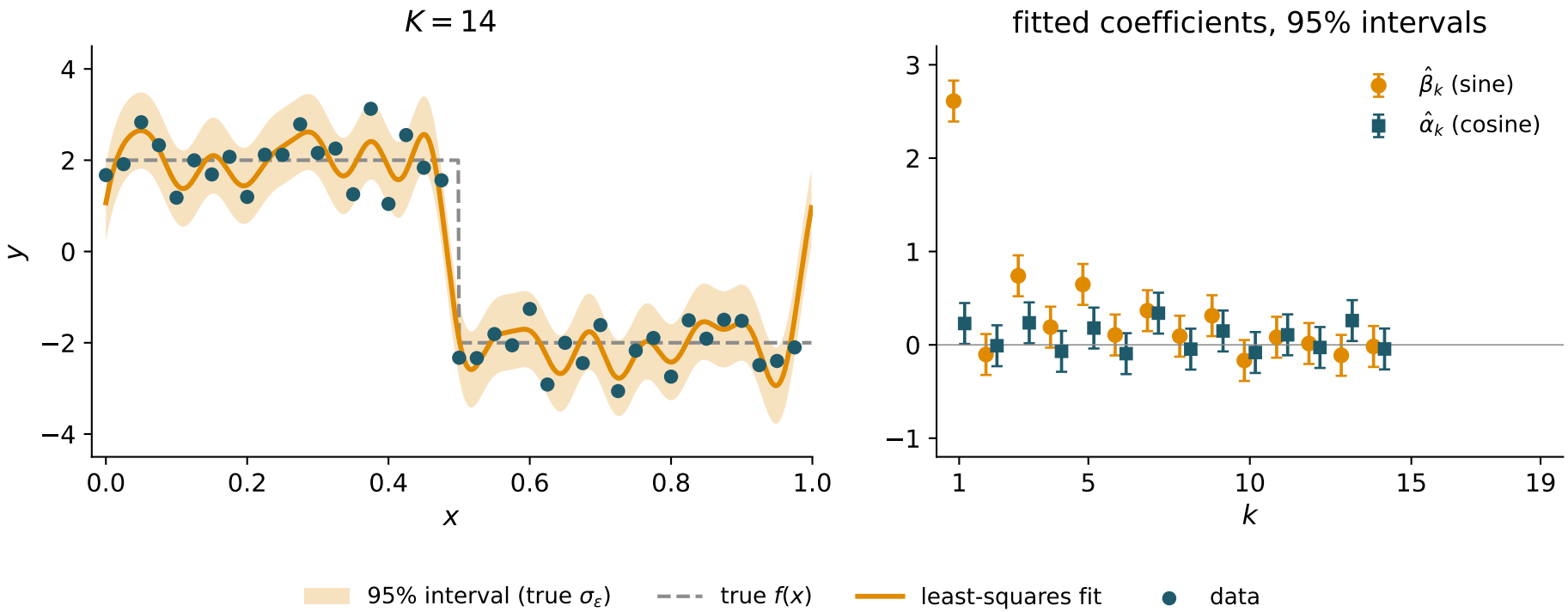
Example: a square wave



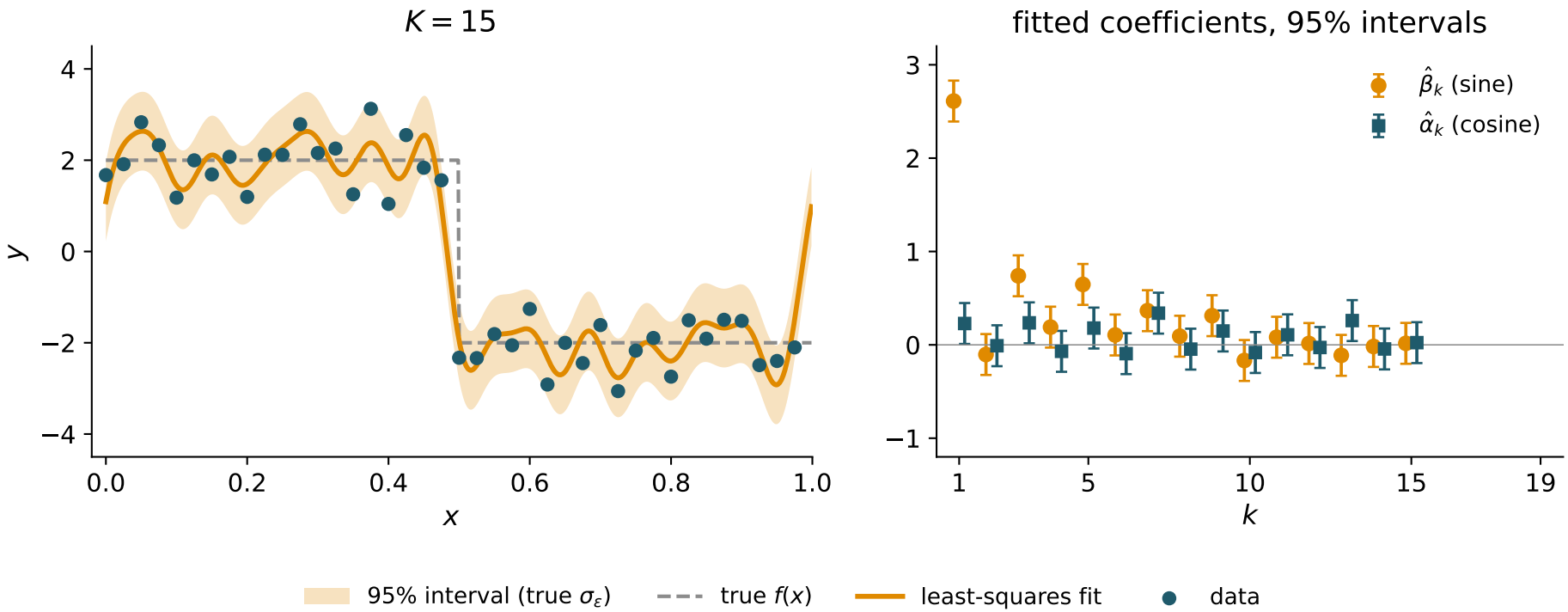
Example: a square wave



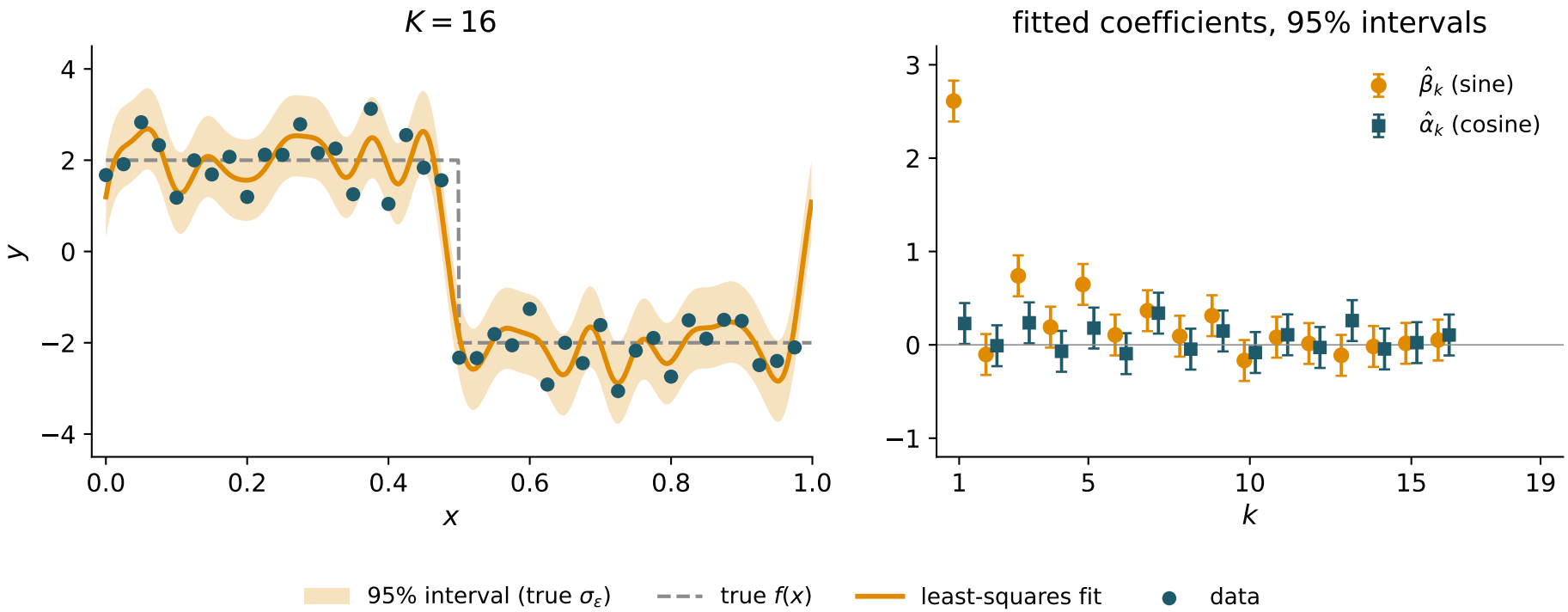
Example: a square wave



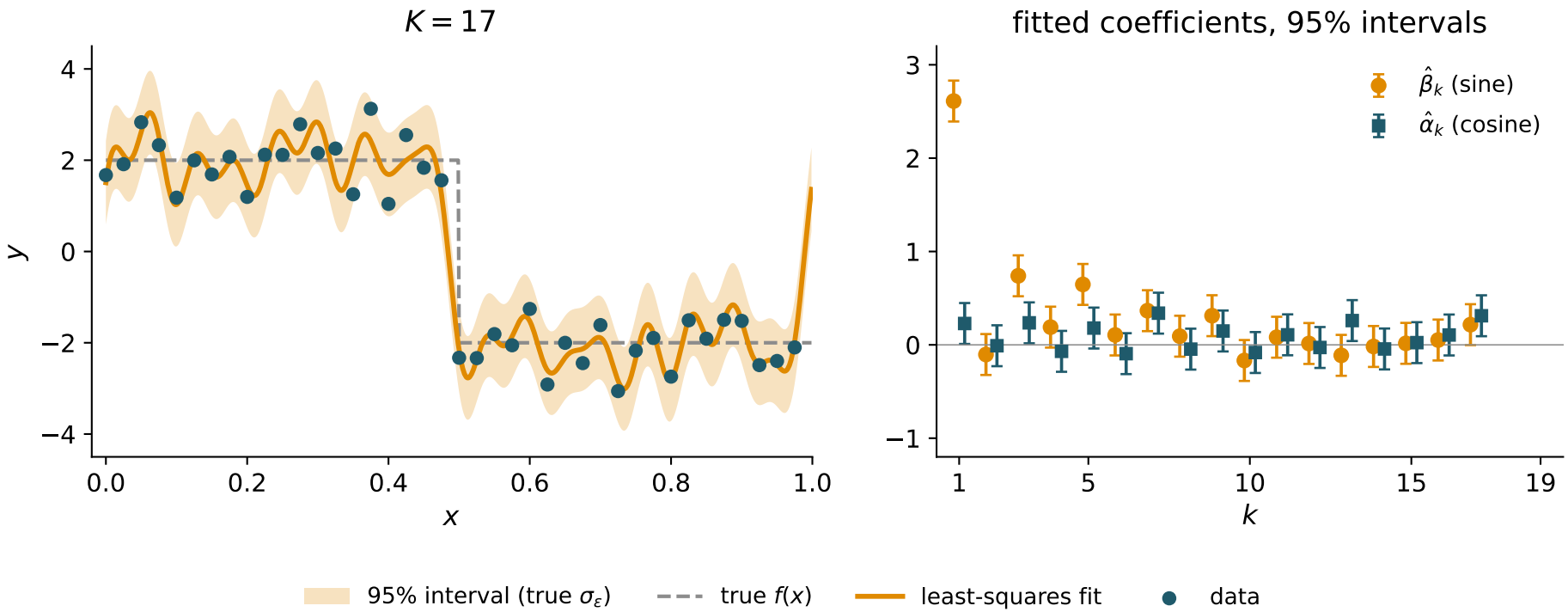
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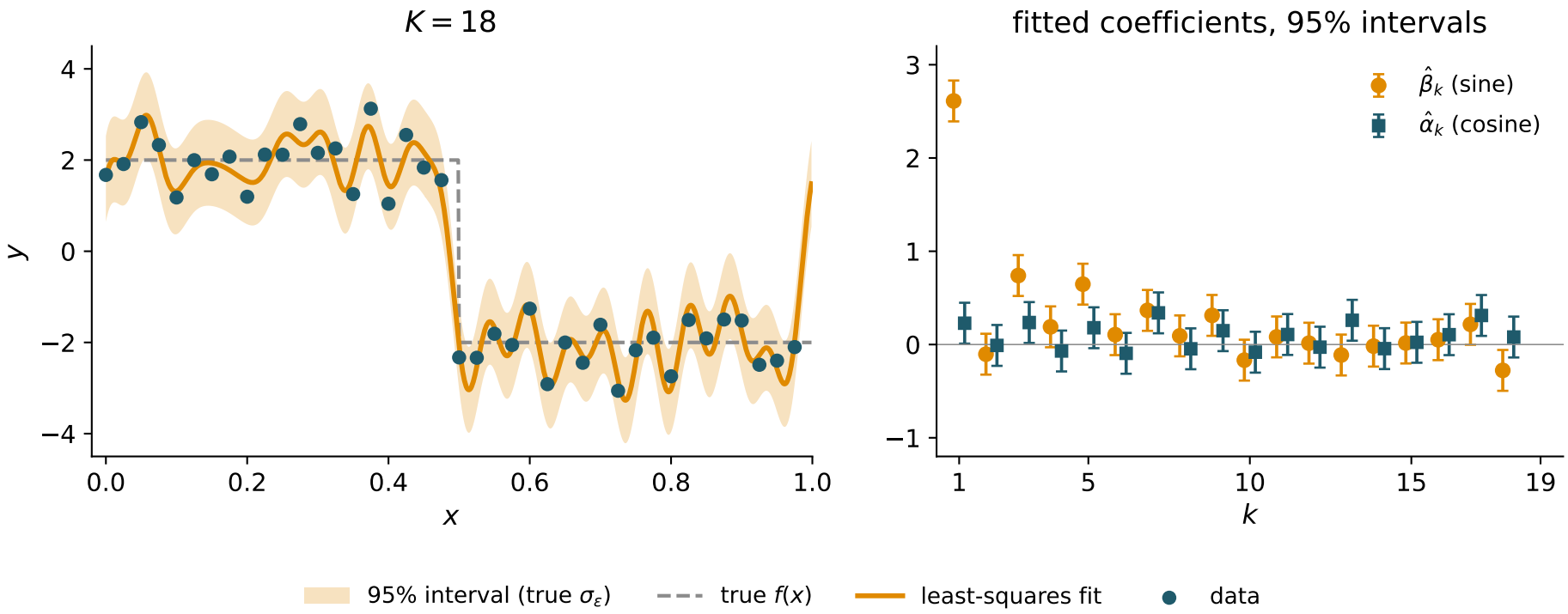
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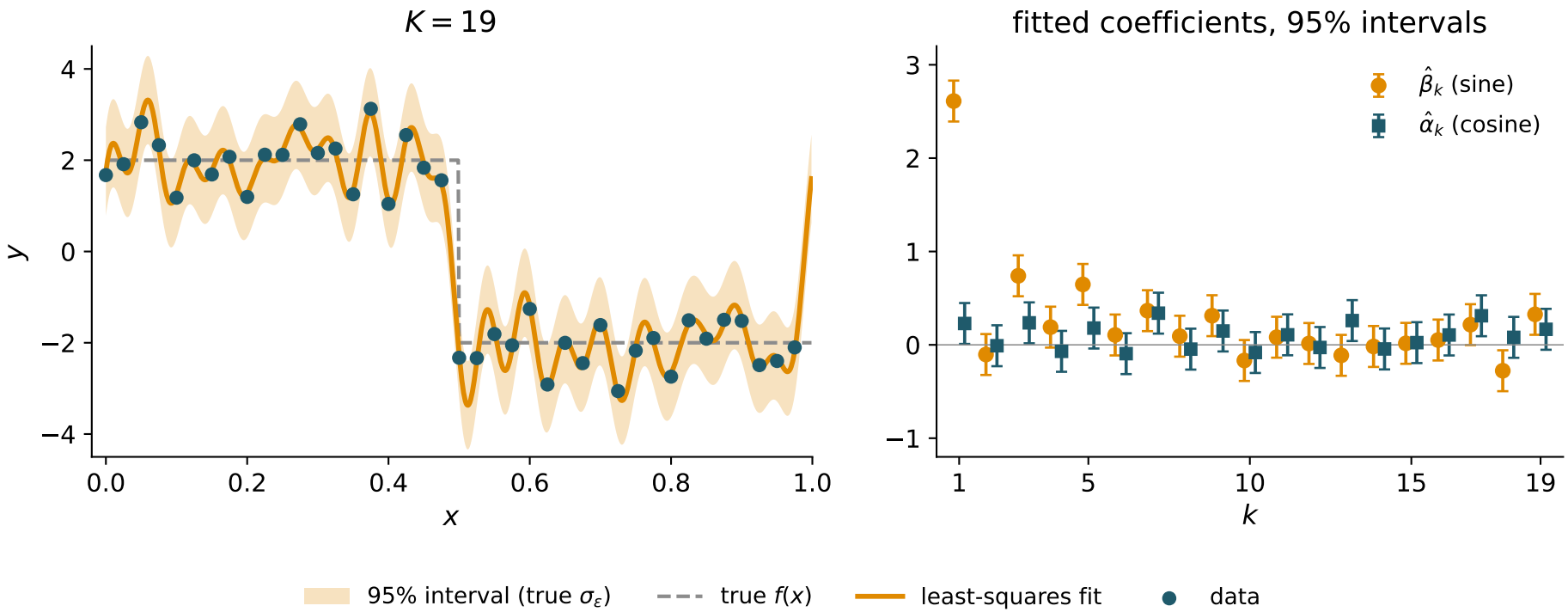
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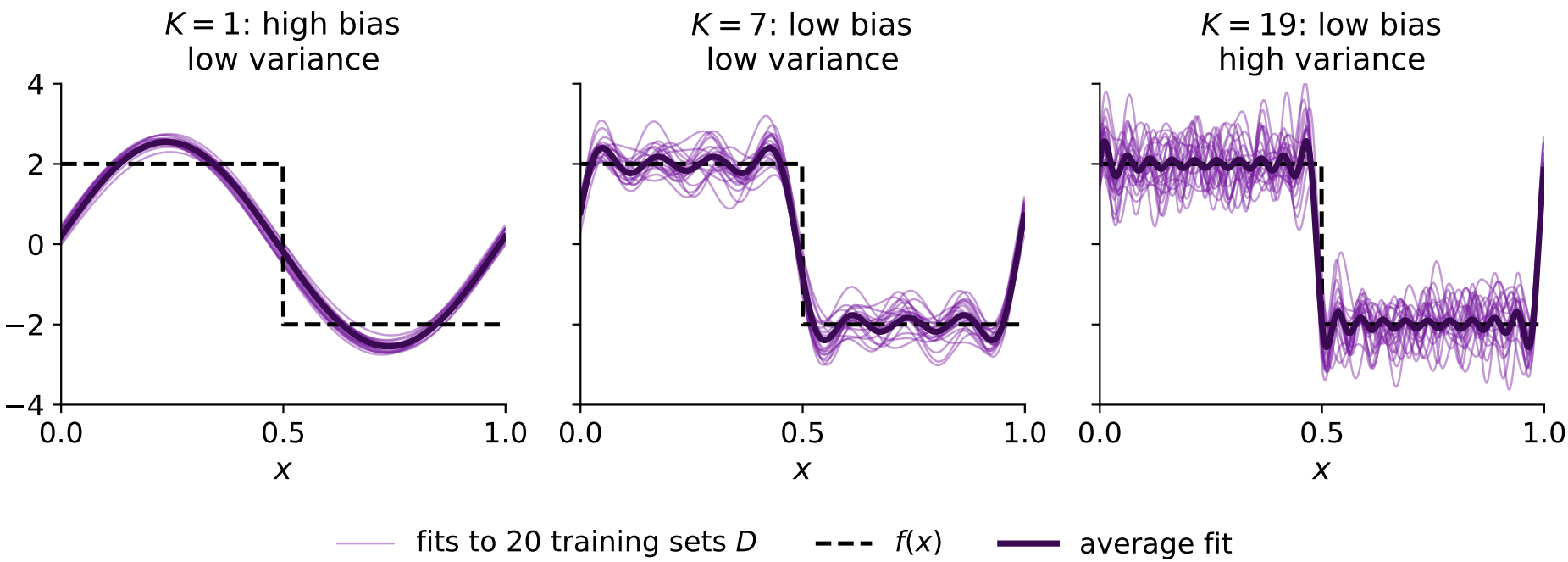
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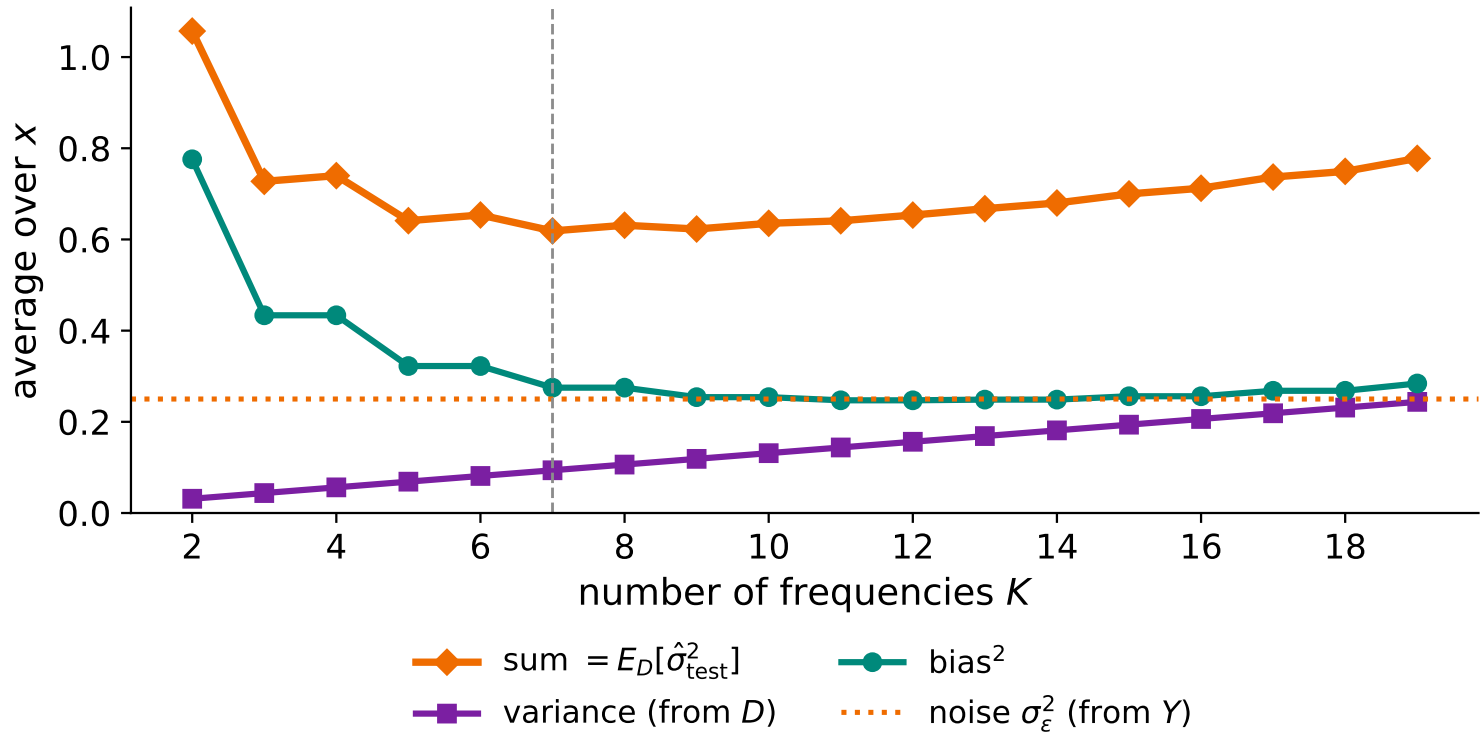
Example: a square wave



Square wave: fits to many training sets



Square wave: bias and variance vs. K



Regularized Fourier coefficients

Now consider the regularized loss for the Fourier model:

$$L_\lambda = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - Y_i\right)^2 + \lambda\sum_{k=1}^{K}\left(\beta_k^2 + \alpha_k^2\right).$$

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Because the features are uncorrelated, the minimizer is again found one coefficient at a time:

$$\hat{\beta}_k^\lambda = \frac{\hat{\beta}_k}{1+2\lambda}, \qquad \hat{\alpha}_k^\lambda = \frac{\hat{\alpha}_k}{1+2\lambda}, \qquad \hat{\beta}_0^\lambda = \overline{Y},$$

where $\hat{\beta}_k$ and $\hat{\alpha}_k$ are the least-squares estimates.

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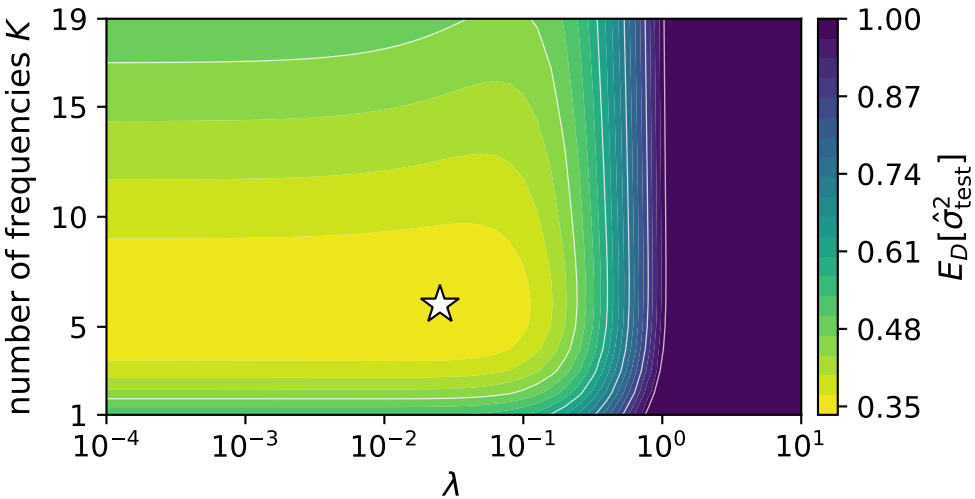
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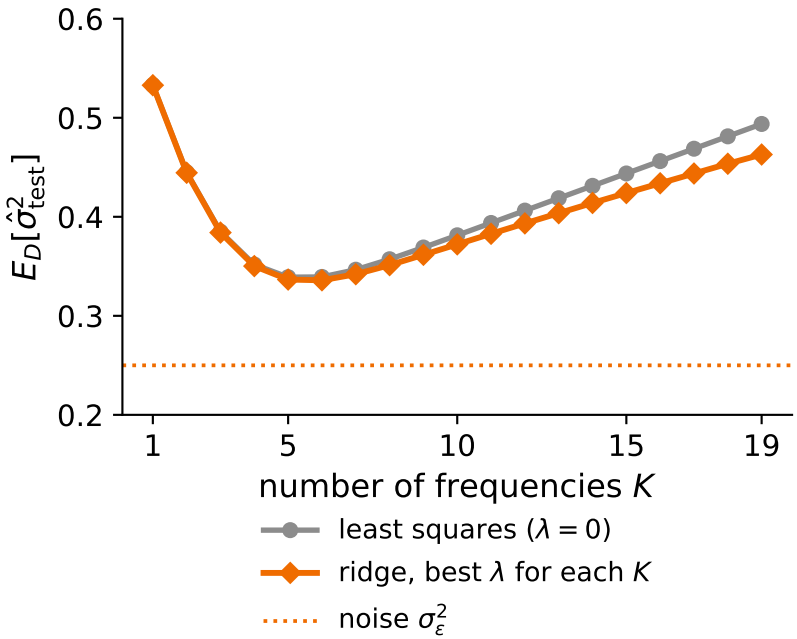
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Each coefficient is the least-squares one shrunk toward 0 by the same factor: $\lambda = 0$ is least squares, and $\lambda \to \infty$ gives $\hat{y} = \overline{Y}$.

Regularized vs. least squares for the Fourier example





Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

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probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

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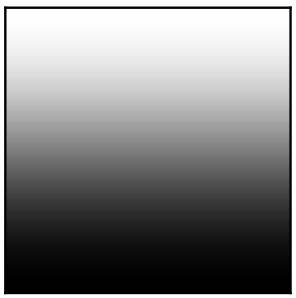
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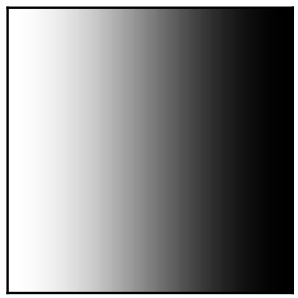
regularization

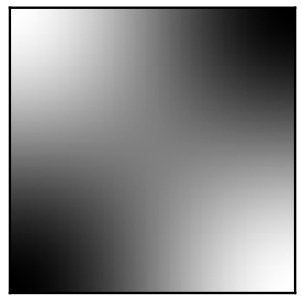
$$L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \sum_{k,l} (k^2 + l^2)\, \beta^2_{k,l}$$

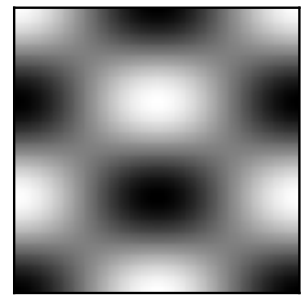
high frequencies cost more

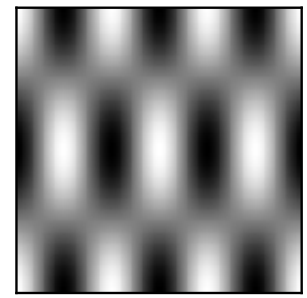
The Fourier features

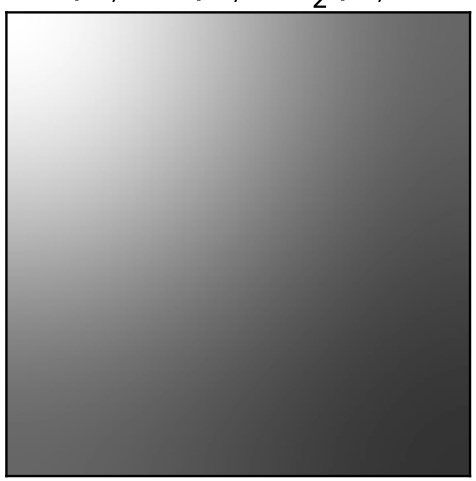
$\phi_{0,1}$ 

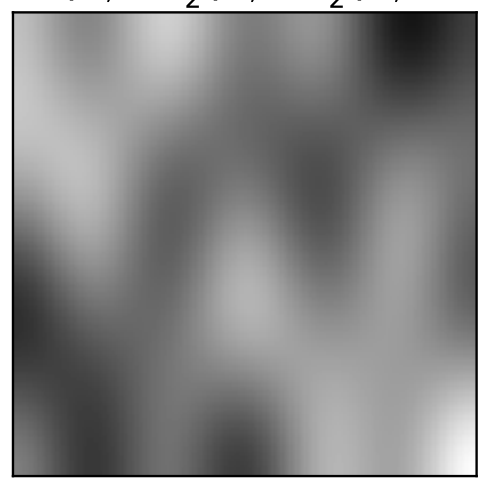
$\phi_{1,0}$ 

$\phi_{1,1}$ 

$\phi_{2,3}$ 

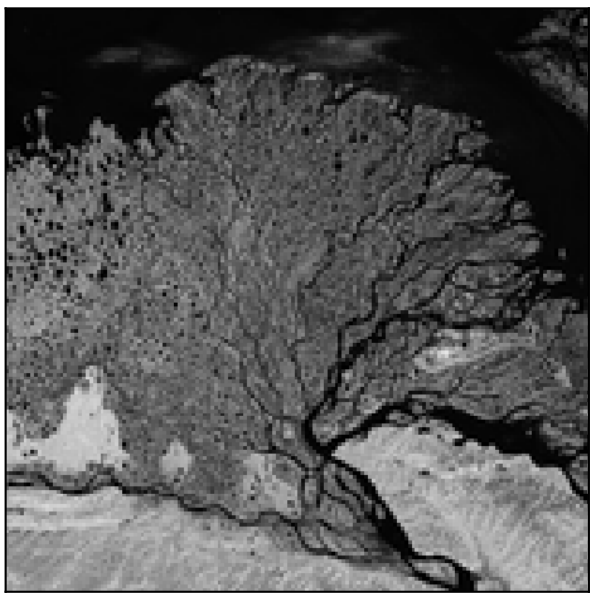
$\phi_{6,2}$ 

$\phi_{0,1} + \phi_{1,0} + \frac{1}{2}\phi_{1,1}$ 

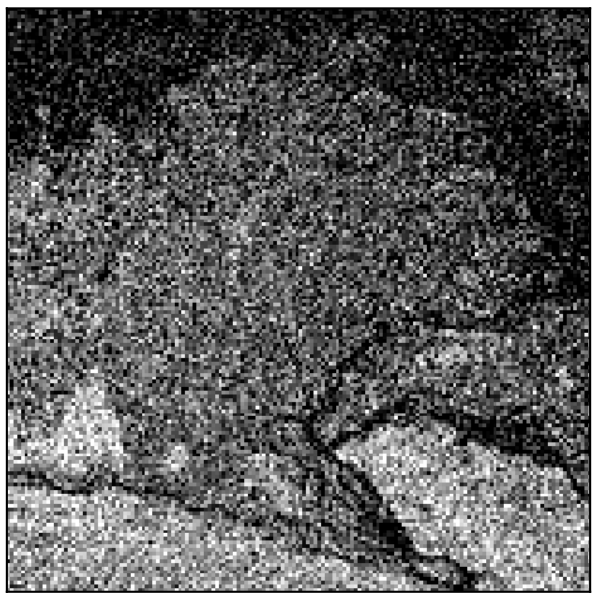
$\phi_{1,1} - \frac{1}{2}\phi_{2,3} + \frac{1}{2}\phi_{6,2}$ 

Smoothing a noisy image

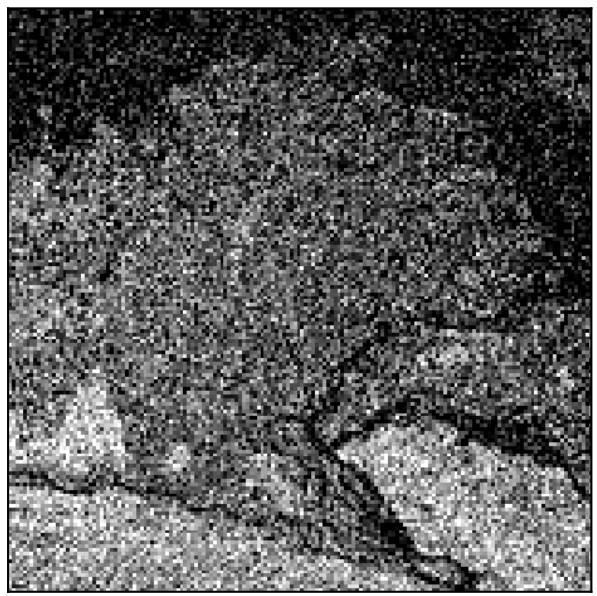
clean image (unknown)



noisy image (data)



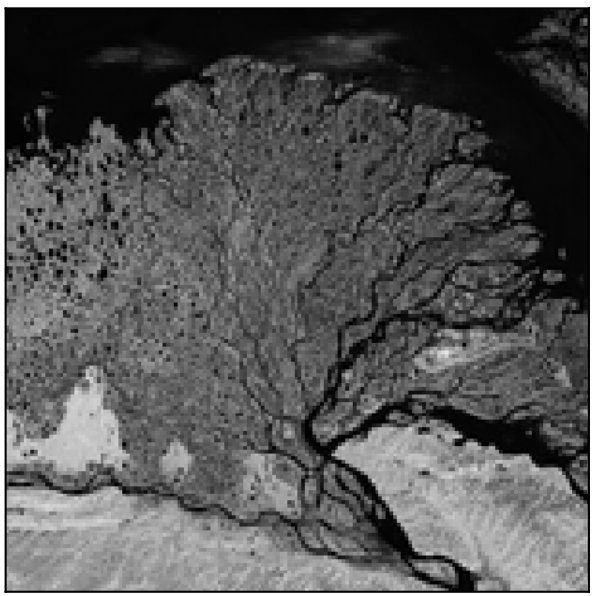
fit, $\lambda = 0$ (least squares)



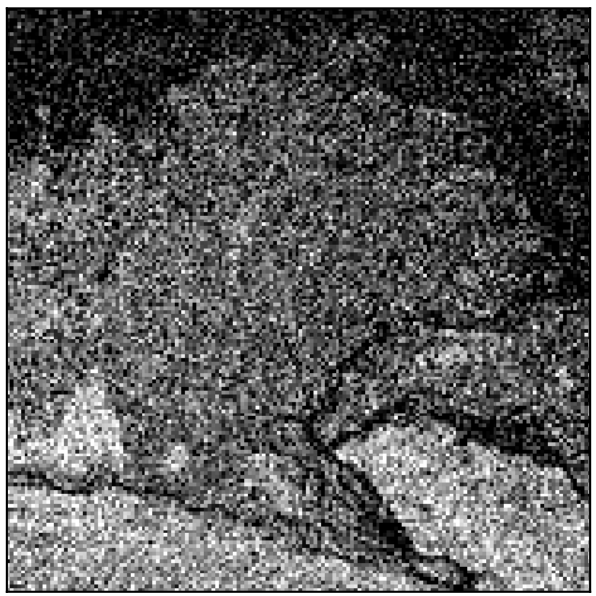
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

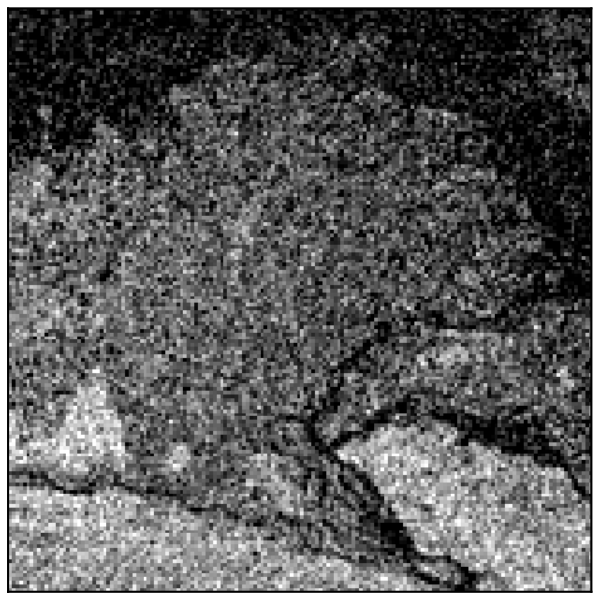
clean image (unknown)



noisy image (data)



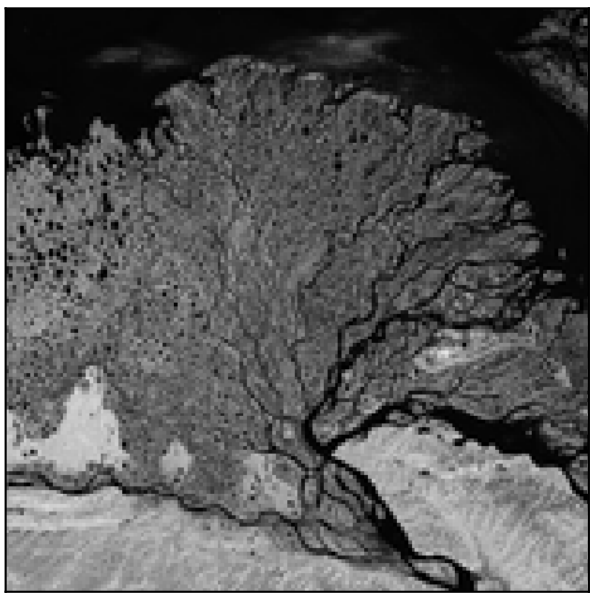
fit, $\lambda = 10^{-9}$



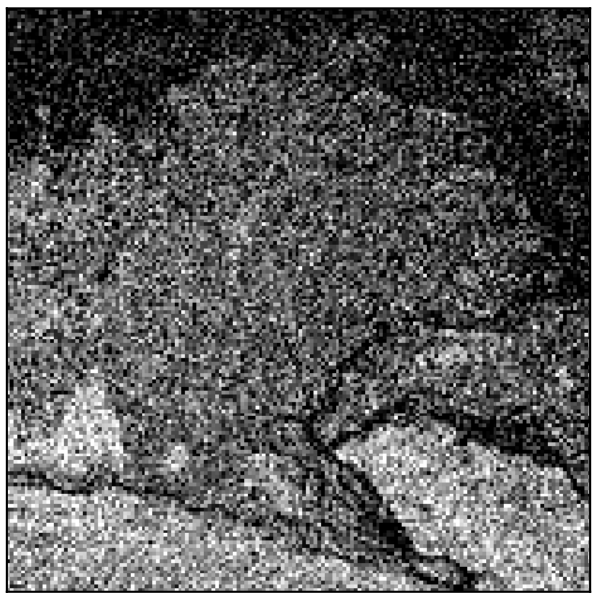
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

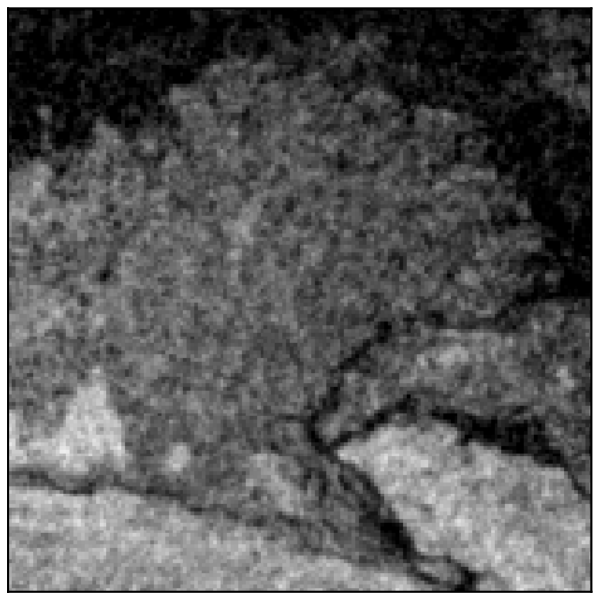
clean image (unknown)



noisy image (data)



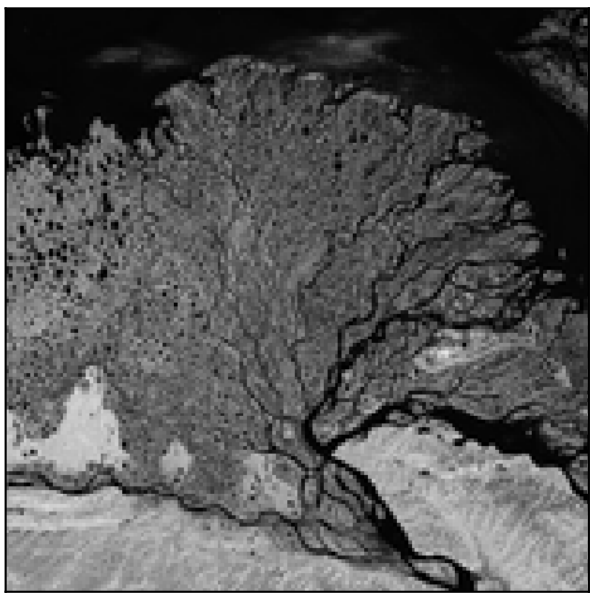
fit, $\lambda = 10^{-8}$



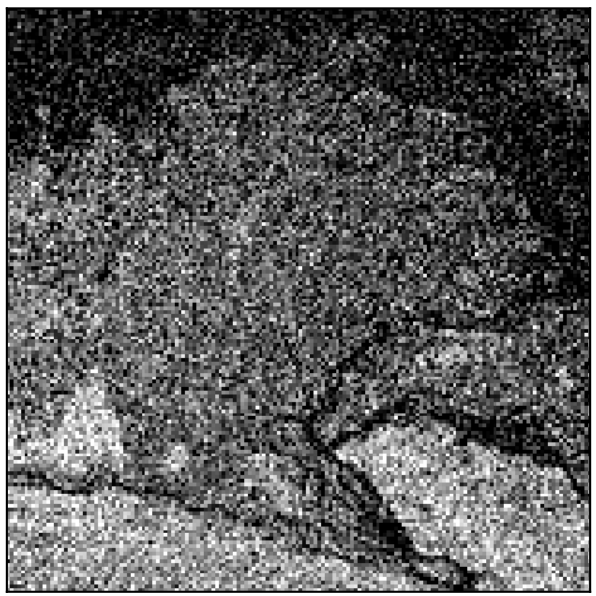
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

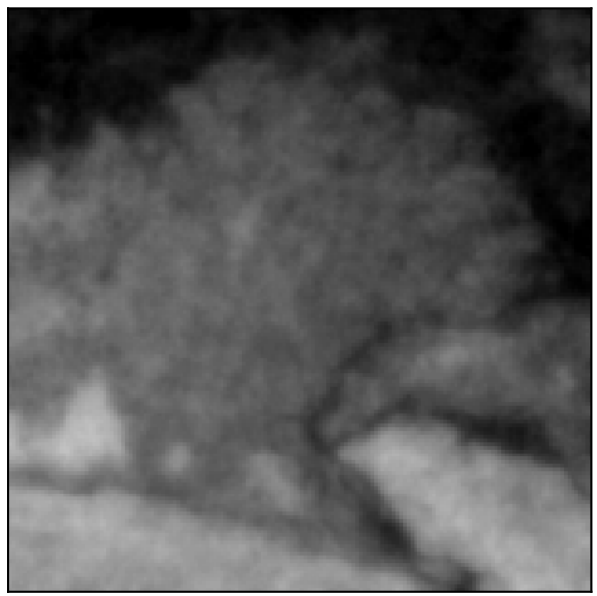
clean image (unknown)



noisy image (data)



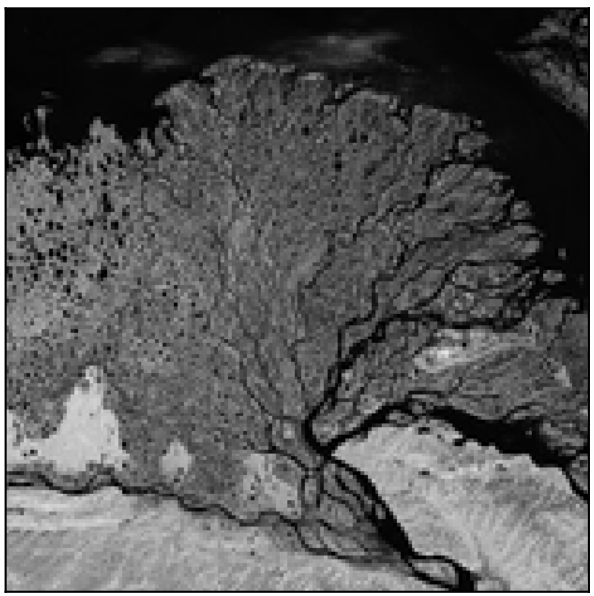
fit, $\lambda = 10^{-7}$



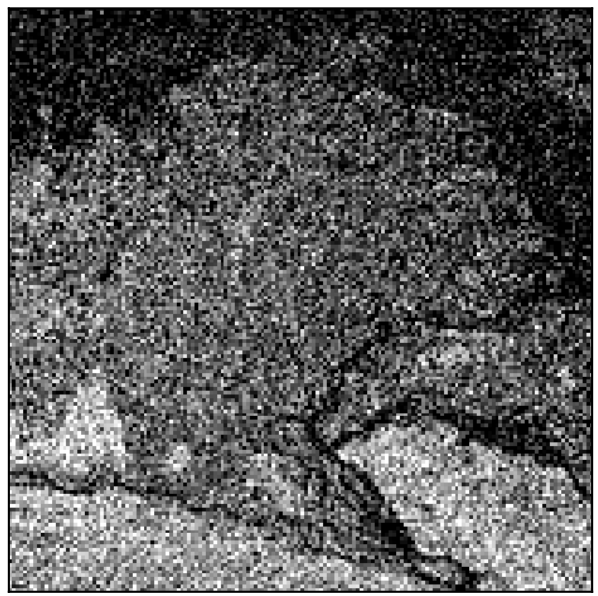
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

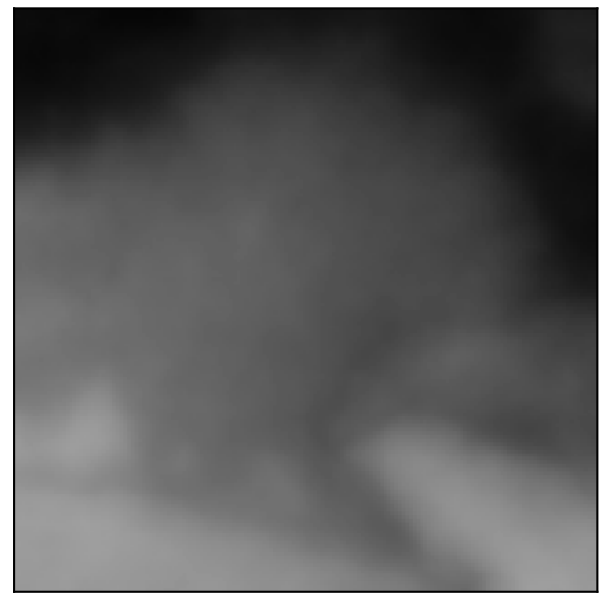
clean image (unknown)



noisy image (data)



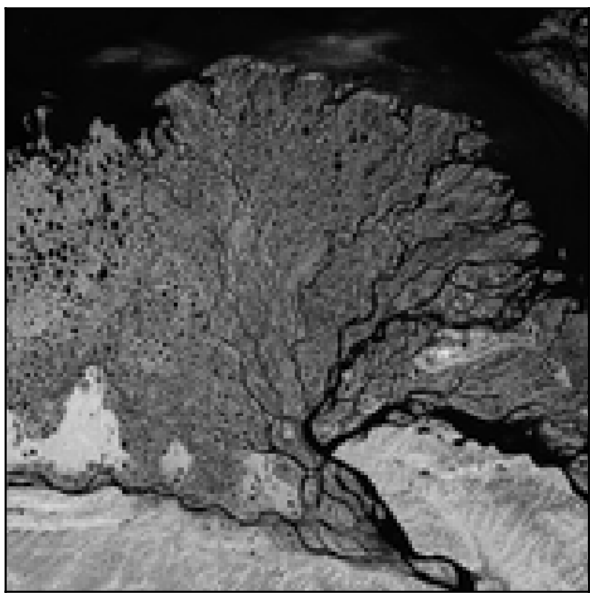
fit, $\lambda = 10^{-6}$



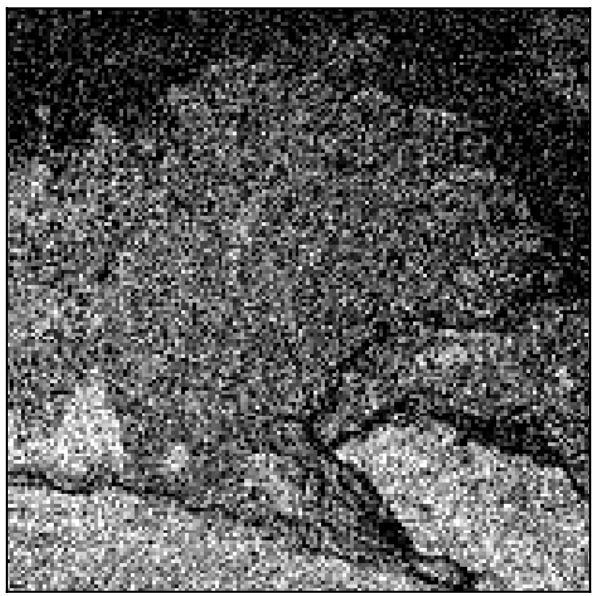
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

clean image (unknown)



noisy image (data)



fit, $\lambda = 10^{-5}$



Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Choosing λ for the image

